NYCU Solid State Physics

Detailed Solutions

Editorial note

These independently derived solutions accompany the supplied Solid State Theory and Solid State Physics examinations. Handwritten responses are used for comparison, not as an answer key. Approximate coefficients, ambiguous wording, missing assumptions, and errors in the reference work are identified where they affect an answer. This is not an official solution set. A quantity that cannot be determined from the surviving problem setup is not assigned an invented value.

Notation and conventions

Unless otherwise stated, the electron spin degeneracy is $g_s = 2$. The particle number is N, the volume is V, and $n = N/V$. The extensive density of states is $D(E)$; $D(E)/V$, or $D(E)/S$ in two dimensions, is the corresponding density per unit volume or area. Spin and valley factors are stated where needed.

Reciprocal vectors satisfy $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$. When only a normal direction is specified, the spacing between successive occupied lattice planes is determined by the shortest nonzero reciprocal vector in that direction.

When a question writes the Coulomb interaction as e^2/r, its unit convention is retained. In SI units this becomes $e^2/(4\pi\epsilon_0 r)$. The symbol Θ denotes the step function. An ordinary stationary point requires a differentiable band with zero gradient; a Dirac band contact is distinguished from an ordinary van Hove stationary point.

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Chapter 1

Solutions: Midterm, 6 May 2014

Only printed Questions 1–3 survive, totaling 60%. They are fully solved here. The final section contains supplementary exercises newly written from topics recognizable in handwritten notes; they are not reconstructions of the missing printed questions. The two-dimensional graphite model in the source is a single honeycomb layer, commonly called graphene.

1. True or false, with assumptions

(a) Identical atoms must be related by a Bravais translation: false

A Bravais translation maps the entire crystal onto itself. Two identical atoms at different basis sites need not be connected by such a translation. Question 3 supplies a counterexample: both basis atoms are carbon, but their separation is

$$\boldsymbol{d}_2 - \boldsymbol{d}_1 = a(0, 1/\sqrt{3}).$$

If this were $n_1\boldsymbol{t}_1 + n_2\boldsymbol{t}_2$ with integers n_i, the x component would give $n_1 = n_2 = n$. The y component would require $\sqrt{3}an = a/\sqrt{3}$, or $n = 1/3$, not an integer.

(b) An exterior point is closer to another reciprocal point: true

The closed first Brillouin zone is

$$\mathcal{B}_1 = \{\boldsymbol{k} : |\boldsymbol{k}| \le |\boldsymbol{k} - \boldsymbol{G}| \text{ for every } \boldsymbol{G} \in \mathcal{L}^*\}.$$

If $\boldsymbol{k} \notin \mathcal{B}_1$, at least one nonzero $\boldsymbol{G}$ violates this inequality: $|\boldsymbol{k} - \boldsymbol{G}| < |\boldsymbol{k}|$. The assertion follows directly from the Wigner–Seitz definition. A boundary point can be equidistant; being strictly outside is different from being on the boundary.

2. FCC plane geometry

The primitive volume and reciprocal basis are

$$\Omega = \frac{a^3}{4}, \qquad \boldsymbol{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \boldsymbol{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \boldsymbol{g}_3 = \frac{2\pi}{a}(1,1,-1).$$

Direct substitution verifies $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$. One lattice point per primitive cell gives volume density $4/a^3$.

(a) Normal $\frac{1}{2}\boldsymbol{g}_1 + \boldsymbol{g}_2 - \boldsymbol{g}_3$

The given vector specifies a direction but is not itself reciprocal. Multiplying by two gives the shortest parallel reciprocal vector:

$$\boldsymbol{G}_0 = \boldsymbol{g}_1 + 2\boldsymbol{g}_2 - 2\boldsymbol{g}_3 = \frac{2\pi}{a}(-1, -3, 5).$$

For $\boldsymbol{R} = \sum_i n_i \boldsymbol{t}_i$,

$$\boldsymbol{G}_0 \cdot \boldsymbol{R} = 2\pi(n_1 + 2n_2 - 2n_3).$$

The integer on the right can take every value m, for instance by setting $n_1 = m$, $n_2 = n_3 = 0$. Consecutive occupied planes are therefore $\boldsymbol{G}_0 \cdot \boldsymbol{r} = 2\pi m$, with

$$\boxed{d = \frac{2\pi}{|\boldsymbol{G}_0|} = \frac{a}{\sqrt{35}}.}$$

Applying $2\pi/|\boldsymbol{G}|$ to the printed half-length normal would skip every other occupied plane.

(b) Surface density

The integer solutions of $n_1 + 2n_2 - 2n_3 = 0$ are generated by

$$\boldsymbol{u} = -2\boldsymbol{t}_1 + \boldsymbol{t}_2, \qquad \boldsymbol{v} = 2\boldsymbol{t}_1 + \boldsymbol{t}_3.$$

Any in-plane point is $n_2\boldsymbol{u} + n_3\boldsymbol{v}$, so these form a primitive planar cell, not a supercell. Its area is

$$A_2 = |\boldsymbol{u} \times \boldsymbol{v}| = \frac{a^2\sqrt{35}}{4}, \qquad \boxed{\rho_2 = \frac{1}{A_2} = \frac{4}{a^2\sqrt{35}}.}$$

Independently, $A_2 d = \Omega$, giving $\rho_2 = (4/a^3)d$. The units are inverse area.

(c) Normal $\boldsymbol{t}_1 + 2\boldsymbol{t}_2 - 2\boldsymbol{t}_3$

Its Cartesian direction is $a(0, -1, 3)/2$. Set a parallel reciprocal vector equal to

$$\boldsymbol{G} = \frac{2\pi}{a}(0, -\lambda, 3\lambda) = h_1\boldsymbol{g}_1 + h_2\boldsymbol{g}_2 + h_3\boldsymbol{g}_3.$$

Solving gives $(h_1, h_2, h_3) = (\lambda, 3\lambda/2, -\lambda/2)$. The smallest positive λ making all coefficients integral is 2:

$$\boldsymbol{G}_0 = \frac{2\pi}{a}(0, -2, 6) = 2\boldsymbol{g}_1 + 3\boldsymbol{g}_2 - \boldsymbol{g}_3, \qquad \boxed{d = \frac{a}{\sqrt{40}}.}$$

Since $\gcd(2, 3, -1) = 1$, every integer plane index occurs and the planes are successive occupied ones.

3. The two honeycomb bands

(a) Check differentiability before classifying

Define

$$x = \frac{ak_x}{2}, \quad y = \frac{\sqrt{3}ak_y}{2}, \quad S = 1 + 4\cos x \cos y + 4\cos^2 x, \quad E_\pm = E_p \pm \gamma\sqrt{S}.$$

For $S > 0$,

$$\nabla E_\pm = \pm\frac{\gamma}{2\sqrt{S}}\nabla S.$$

At $S = 0$, this formula cannot simply be evaluated; expand the original dispersion.

$\Gamma = (0,0)$**: maximum and minimum.** Expanding the cosines gives

$$S = 9 - \frac{3}{2}a^2q^2 + O(q^4), \qquad E_\pm = E_p \pm 3\gamma \mp \frac{\gamma a^2}{4}q^2 + O(q^4).$$

The gradient vanishes and the Hessian is $\mp\gamma a^2 I/2$. Thus the upper band has a maximum $E_p + 3\gamma$, and the lower band a minimum $E_p - 3\gamma$.

$Q = (\pi/a, \pi/(\sqrt{3}a))$**: saddle.** Write $\boldsymbol{k} = \boldsymbol{Q} + \boldsymbol{q}$, $u = aq_x/2$, $v = \sqrt{3}aq_y/2$. Since $x = \pi/2 + u$ and $y = \pi/2 + v$,

$$S = 1 + 4u^2 + 4uv + O(q^4), \qquad \sqrt{S} = 1 + 2u^2 + 2uv + O(q^4).$$

Consequently

$$E_\pm = E_p \pm \gamma \pm \frac{\gamma a^2}{2}(q_x^2 + \sqrt{3}q_xq_y) + O(q^4).$$

The upper-band Hessian is

$$\gamma a^2 \begin{pmatrix} 1 & \sqrt{3}/2 \\ \sqrt{3}/2 & 0 \end{pmatrix},$$

with eigenvalues $3\gamma a^2/2$ and $-\gamma a^2/2$. Both bands are saddles. In rotated principal-axis coordinates,

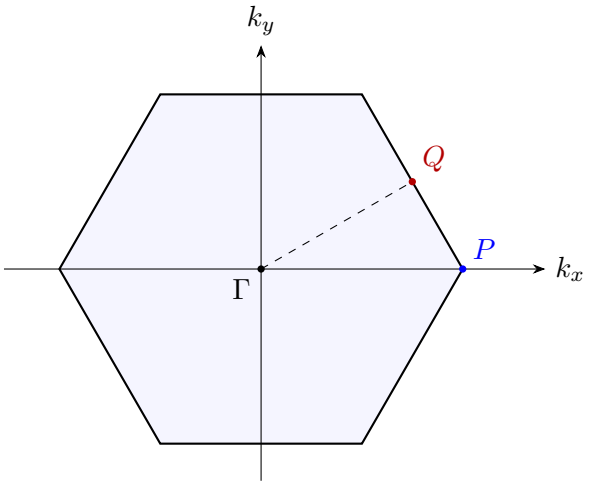
$$E_+ - (E_p + \gamma) = \alpha q_1^2 - \eta q_2^2 + O(q^4), \qquad \alpha = \frac{3\gamma a^2}{4}, \quad \eta = \frac{\gamma a^2}{4}.$$

The lower band reverses both signs and remains a saddle.

$P = (4\pi/(3a), 0)$**: Dirac touching.** Here $S = 0$. Expanding gives

$$S = \frac{3}{4}a^2q^2 + O(q^3), \qquad E_\pm = E_p \pm \frac{\sqrt{3}\gamma a}{2}|\boldsymbol{q}| + O(q^2).$$

Every line through the apex has an absolute-value cusp. The band is therefore **not differentiable** at the touching. If a critical point means a differentiable point with $\nabla E = 0$, P is not one. Under a broader definition allowing nonsmooth extrema, the upper and lower bands have conical minimum and maximum, respectively. One must not infer $\nabla E = 0$ from $\nabla S = 0$ at $S = 0$.



First Brillouin zone, in units of $2\pi/a$. The source's P, Q are commonly called K, M.

(b) There is no gap

The band separation is $E_+ - E_- = 2\gamma|F_{\boldsymbol{k}}| \geq 0$ and vanishes at P and symmetry-equivalent corners:

$$\boxed{E_g = 0.}$$

At half filling with $E_F = E_p$, the Fermi surface reduces to Dirac points in this model.

(c) Choose the lower-band minimum at Γ

Use DOS per area, per spin, for one local patch:

$$D(E) = \int_{\text{patch}} \frac{\mathrm{d}^2 q}{(2\pi)^2} \delta(E - E(\boldsymbol{q})).$$

Let $E_0 = E_p - 3\gamma$, $A = \gamma a^2/4$, and approximate $E = E_0 + Aq^2$ in a circular patch $q < \Lambda$. Setting $z = Aq^2$ gives $q\,\mathrm{d}q = \mathrm{d}z/(2A)$:

$$\begin{aligned} D_\Gamma(E) &= \frac{1}{2\pi} \int_0^\Lambda q\,\mathrm{d}q\,\delta(E - E_0 - Aq^2) \\ &= \boxed{\frac{1}{\pi\gamma a^2}\Theta(E - E_0)\Theta(E_0 + A\Lambda^2 - E).} \end{aligned}$$

The local approximation gives a constant step at the band edge; the upper cutoff depends on the patch. Multiply by two for both spins and by sample area $\mathcal{A}$ for a total DOS.

Additional check: the saddle at Q. For the upper band set $\varepsilon = E - (E_p + \gamma)$. With $\varepsilon = \alpha q_1^2 - \eta q_2^2$, use $u = \sqrt{\alpha} q_1$, $v = \sqrt{\eta} q_2$, then $s = u + v$, $t = u - v$. Thus $u^2 - v^2 = st$ and $\mathrm{d}u\,\mathrm{d}v = \mathrm{d}s\,\mathrm{d}t/2$. For a local cutoff $|s|, |t| < L$,

$$\begin{aligned} D_Q(E) &\simeq \frac{1}{8\pi^2\sqrt{\alpha\eta}} \int_{-L}^{L} \mathrm{d}s \int_{-L}^{L} \mathrm{d}t\,\delta(\varepsilon - st) \\ &= \frac{1}{4\pi^2\sqrt{\alpha\eta}} \ln\frac{L^2}{|\varepsilon|} \\ &= \frac{1}{\pi^2\sqrt{3}\gamma a^2} \ln\frac{E_c}{|E - (E_p + \gamma)|}. \end{aligned}$$

This logarithmic van Hove divergence is per saddle and per spin. Its cutoff E_c cannot be fixed by the quadratic expansion alone. In contrast, one Dirac valley gives

$$D_P(E) = \frac{|E - E_p|}{2\pi\beta^2}, \qquad \beta = \sqrt{3}\gamma a/2,$$

which vanishes at the touching. These are three distinct DOS signatures.

Supplementary exercises from handwritten topics

The topics below are recognizable in attached handwritten answers, but their printed question statements, weights, and exact requested scope are missing. Exercises A–D are newly composed learning material, not verbatim reconstructions of original Questions 4–7.

A. Derive Bloch form from translation symmetry

Exercise. For a Hamiltonian with Bravais translation symmetry, derive the quasiperiodicity of a simultaneous energy and translation eigenstate.

Define $(T_{\boldsymbol{R}}\psi)(\boldsymbol{r}) = \psi(\boldsymbol{r}+\boldsymbol{R})$. Then $[H, T_{\boldsymbol{R}}] = 0$ and $T_{\boldsymbol{R}}T_{\boldsymbol{R}'} = T_{\boldsymbol{R}+\boldsymbol{R}'}$. Choose a simultaneous eigenstate within each energy eigenspace:

$$T_{\boldsymbol{R}}\psi = \lambda(\boldsymbol{R})\psi.$$

Translations are unitary, so $|\lambda| = 1$. The group law imposes $\lambda(\boldsymbol{R}+\boldsymbol{R}') = \lambda(\boldsymbol{R})\lambda(\boldsymbol{R}')$. Choose the primitive phases as $\lambda(\boldsymbol{t}_i) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{t}_i}$; then

$$\psi(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r}).$$

Defining $u_{\boldsymbol{k}} = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi$ gives $u(\boldsymbol{r}+\boldsymbol{R}) = u(\boldsymbol{r})$. Vectors $\boldsymbol{k}$ and $\boldsymbol{k}+\boldsymbol{G}$ have the same translation eigenvalues, so $\boldsymbol{k}$ may be chosen in the first Brillouin zone.

B. Brillouin's theorem and single excitations

Exercise. Show that a self-consistent HF determinant has zero Hamiltonian matrix element with every singly excited determinant.

Let Φ_0 be the stationary HF determinant and Φ_i^a replace occupied spin orbital i by virtual orbital a. The Slater–Condon rules give

$$\langle\Phi_i^a|H|\Phi_0\rangle = h_{ai} + \sum_{j\in\mathrm{occ}}\big(\langle aj|v|ij\rangle - \langle aj|v|ji\rangle\big) = F_{ai}.$$

For canonical orbitals, $F\chi_i = \epsilon_i\chi_i$ and $\langle\chi_a \mid \chi_i\rangle = 0$, hence

$$\boxed{\langle\Phi_i^a|H|\Phi_0\rangle = 0.}$$

Equivalently, varying the real and imaginary components of the orbital rotation $\chi_i \mapsto \chi_i + \theta\chi_a$ must give zero first-order energy change. The theorem removes only the coupling to single excitations; double-excitation couplings generally survive, so HF need not be exact.

C. A determinant is an exact noninteracting eigenstate

Exercise. If $H_0 = \sum_{j=1}^{N} h(j)$ and $h\chi_i = \epsilon_i\chi_i$, show that the determinant has energy $\sum_i \epsilon_i$.

Expand the determinant:

$$\Phi = \frac{1}{\sqrt{N!}}\sum_P (-1)^P \prod_{j=1}^{N}\chi_{P(j)}(x_j).$$

For every permutation, H_0 acting on the product gives the same eigenvalue $\sum_j \epsilon_{P(j)} = \sum_i \epsilon_i$. Therefore

$$H_0\Phi = \left(\sum_{i=1}^{N}\epsilon_i\right)\Phi.$$

The ground state fills the lowest available N spin orbitals. Adding a true two-body interaction invalidates this argument, which is why an exact noninteracting determinant must be distinguished from an interacting HF approximation.

D. Hartree, Hartree–Fock, and Kohn–Sham

Exercise. Compare the single-particle equations, variational objects, and treatment of exchange and correlation.

Let $h = -\hbar^2\nabla^2/(2m_e) + v_{\text{ext}}$ and $\kappa = e^2/(4\pi\epsilon_0)$.

(i) **Hartree.** Vary a product $\prod_i \chi_i$:

$$\left[h + \sum_{j\neq i} J_j\right]\chi_i = \epsilon_i\chi_i, \qquad J_j(\boldsymbol{r}) = \int \frac{\kappa|\chi_j(x')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'.$$

It includes the other electrons' average Coulomb field but no antisymmetry-driven exchange.

(ii) **Hartree–Fock.** Vary a single determinant:

$$\left[h + \sum_j (J_j - K_j)\right]\chi_i = \epsilon_i\chi_i,$$

$$(K_j\chi_i)(x) = \chi_j(x)\int \frac{\kappa\chi_j^*(x')\chi_i(x')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'.$$

Exchange is nonlocal and $J_i - K_i$ cancels self-interaction. Correlation beyond a single determinant is generally missing.

(iii) **Kohn–Sham.** Introduce auxiliary noninteracting orbitals reproducing the interacting density:

$$F[n] = T_s[n] + E_H[n] + E_{xc}[n], \quad E_H = \frac{\kappa}{2}\iint \frac{n(\boldsymbol{r})n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r\,\mathrm{d}^3r'.$$

Orbital variation gives

$$[h + v_H(\boldsymbol{r}) + v_{xc}(\boldsymbol{r})]\varphi_i = \epsilon_i\varphi_i, \qquad n = \sum_i f_i|\varphi_i|^2,$$

with spin summation understood, $v_H = \delta E_H/\delta n$, and $v_{xc} = \delta E_{xc}/\delta n$. The functional E_{xc} contains both nonclassical interactions and the difference between true and auxiliary kinetic energies. It is not merely HF exchange renamed. An exact functional gives exact ground-state density and energy in principle; approximate functionals add their own errors.

Primary references used for checks

P. Hohenberg and W. Kohn, Inhomogeneous Electron Gas, Phys. Rev. 136, B864 (1964); W. Kohn and L. J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, Phys. Rev. 140, A1133 (1965). Exercises A–D are explicitly new exercises based on the note topics. The original printed continuation remains missing.

Chapter 2

Solutions: Midterm, 5 December 2014

All seven printed questions survive. Their weights total 110% and are retained. The derivations below are independent of handwritten answers. In Question 2, a is the conventional cubic lattice parameter; in Question 3 it is the two-dimensional primitive-translation length.

1. Eight definitions

(a) **Bravais lattice.** A three-dimensional Bravais lattice is the point set

$$\mathcal{L} = \{\boldsymbol{R} = n_1\boldsymbol{t}_1 + n_2\boldsymbol{t}_2 + n_3\boldsymbol{t}_3 : n_i \in \mathbb{Z}\},$$

with linearly independent primitive vectors. Every lattice point has the same environment under translation. A lattice describes translational symmetry; a lattice point need not represent one atom.

(b) **Composite crystal.** In the lattice-plus-basis terminology relevant here, attach the same atomic basis $\{(\boldsymbol{d}_\alpha, s_\alpha)\}$ to each Bravais point. Positions are $\boldsymbol{R} + \boldsymbol{d}_\alpha$, with s_α specifying species. Even identical atoms at different basis sites need not be related by a Bravais translation. The honeycomb crystal has a two-atom basis on a triangular Bravais lattice.

(c) **Brillouin zone.** The first Brillouin zone is the reciprocal origin's Wigner–Seitz cell:

$$|\boldsymbol{k}| \le |\boldsymbol{k} - \boldsymbol{G}| \quad \text{for every } \boldsymbol{G} \in \mathcal{L}^*.$$

Its faces are perpendicular bisectors between the origin and other reciprocal points. Higher zones can be classified by the number of crossed Bragg planes.

(d) **Reciprocal lattice.** With $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$,

$$\mathcal{L}^* = \{m_1\boldsymbol{g}_1 + m_2\boldsymbol{g}_2 + m_3\boldsymbol{g}_3 : m_i \in \mathbb{Z}\}.$$

Equivalently $\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ for every direct-lattice translation.

(e) **Bloch theorem.** For a single-particle Hamiltonian with lattice symmetry, eigenstates can be chosen as

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}(\boldsymbol{r}), \quad u_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}).$$

Thus $\psi(\boldsymbol{r} + \boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi(\boldsymbol{r})$: ψ is generally quasiperiodic, while u is periodic. The theorem follows from simultaneous diagonalization of the Hamiltonian and commuting translations.

(f) **Exchange energy.** For a determinant of orthonormal spin orbitals $\chi_i(x)$, $x = (\boldsymbol{r}, \sigma)$,

$$E_{ee} = \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}), \qquad E_x = -\frac{1}{2}\sum_{ij} K_{ij},$$

$$K_{ij} = \iint \chi_i^*(x)\chi_j^*(x')\frac{\kappa\,\chi_j(x)\chi_i(x')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}x\,\mathrm{d}x', \qquad \kappa = \frac{e^2}{4\pi\epsilon_0}.$$

Exchange arises from antisymmetry; separated spin functions restrict it to equal-spin pairs. It is not an additional classical force or the entire correlation energy.

(g) **Fully spin-polarized ground state.** All spins point in the same direction: $N_\uparrow = N$, $N_\downarrow = 0$ gives $\zeta = (N_\uparrow - N_\downarrow)/N = 1$, $S_z = N\hbar/2$, and total-spin quantum number $S = N/2$. It does not mean zero spin.

(h) **Wigner crystallization.** At sufficiently low density, an electron gas in a uniform positive background can form an ordered electron crystal. With mean spacing $r_s a_0$, kinetic and Coulomb scales vary as r_s^{-2} and r_s^{-1}, respectively; interactions become relatively more important at large r_s. This is spontaneous density ordering, not merely drawing a Wigner–Seitz cell.

2. Lattice planes and translation phase

Reciprocal basis and primitive volume

From $\boldsymbol{g}_1 = 2\pi(\boldsymbol{t}_2 \times \boldsymbol{t}_3)/\Omega$ and cyclic permutations,

$$\Omega = \boldsymbol{t}_1 \cdot (\boldsymbol{t}_2 \times \boldsymbol{t}_3) = \frac{a^3}{4},$$

$$\boxed{\boldsymbol{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \boldsymbol{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \boldsymbol{g}_3 = \frac{2\pi}{a}(1,1,-1).}$$

One point per primitive cell gives $\rho_3 = 4/a^3$. For coprime integers h_i, let $\boldsymbol{G}_0 = \sum_i h_i \boldsymbol{g}_i$. At lattice points,

$$\boldsymbol{G}_0 \cdot \boldsymbol{R} = 2\pi(h_1 n_1 + h_2 n_2 + h_3 n_3) = 2\pi m.$$

Every integer m occurs. Thus successive planes containing lattice points have

$$\boxed{d = 2\pi/|\boldsymbol{G}_0|.}$$

Use the shortest reciprocal vector parallel to the specified normal; otherwise the level planes may include empty planes.

(a) Normal $3\boldsymbol{g}_1 + 6\boldsymbol{g}_2 - 6\boldsymbol{g}_3$

Remove the common factor 3:

$$\boldsymbol{G}_0 = \boldsymbol{g}_1 + 2\boldsymbol{g}_2 - 2\boldsymbol{g}_3 = \frac{2\pi}{a}(-1,-3,5), \qquad \gcd(1,2,-2) = 1.$$

Consequently

$$\boxed{d = a/\sqrt{35}.}$$

Directly using the printed normal's length would incorrectly divide this answer by three.

(b) Surface density

A primitive planar cell of area A_2, stacked by a lattice translation with normal component d, gives $A_2 d = \Omega$. Hence

$$\boxed{\rho_2 = \frac{1}{A_2} = \frac{d}{\Omega} = \frac{4}{a^2\sqrt{35}}.}$$

For an explicit check, all integer solutions of $n_1 + 2n_2 - 2n_3 = 0$ are generated by

$$\boldsymbol{u} = -2\boldsymbol{t}_1 + \boldsymbol{t}_2 = \frac{a}{2}(1, -2, -1), \qquad \boldsymbol{v} = 2\boldsymbol{t}_1 + \boldsymbol{t}_3 = \frac{a}{2}(1, 3, 2).$$

Their cross-product magnitude is $A_2 = a^2\sqrt{35}/4$.

(c) Normal $\frac{1}{2}\boldsymbol{t}_1 + \boldsymbol{t}_2 - \boldsymbol{t}_3$

The direction is $a(0, -1, 3)/4$. A reciprocal vector has components

$$\frac{2\pi}{a}(-h_1 + h_2 + h_3,\ h_1 - h_2 + h_3,\ h_1 + h_2 - h_3).$$

Setting these equal to $(2\pi/a)(0, -\lambda, 3\lambda)$ gives $(h_1, h_2, h_3) = (\lambda, 3\lambda/2, -\lambda/2)$. The smallest positive λ giving integer coefficients is 2:

$$\boldsymbol{G}_0 = 2\boldsymbol{g}_1 + 3\boldsymbol{g}_2 - \boldsymbol{g}_3 = \frac{2\pi}{a}(0, -2, 6), \qquad \boxed{d = a/\sqrt{40}.}$$

The shorter vector $(2\pi/a)(0, -1, 3)$ is not reciprocal: its primitive coefficients are half-integers.

(d) Periodicity

For $f(\boldsymbol{r}) = \exp(\mathrm{i}\boldsymbol{g}_1 \cdot \boldsymbol{r}/2)$,

$$f(\boldsymbol{r} + \boldsymbol{t}_1) = f(\boldsymbol{r})\mathrm{e}^{\mathrm{i}\pi} = -f(\boldsymbol{r}).$$

Thus **no**. The squared magnitude is uniform, but the complex wavefunction retains only translations with even n_1, not all periods of the original lattice.

3. Critical points and two-dimensional DOS

Let $E_\pm = E_p \pm \gamma\sqrt{S}$, with

$$S = 1 + 4\cos x \cos y + 4\cos^2 x, \quad x = \frac{ak_x}{2}, \quad y = \frac{\sqrt{3}ak_y}{2}.$$

An ordinary stationary critical point is differentiable and satisfies $\nabla E = 0$. At a touching, $\nabla S = 0$ does not by itself imply $\nabla\sqrt{S} = 0$.

(a) Γ and P

At Γ,

$$S = 9 - 6x^2 - 2y^2 + O(k^4) = 9 - \frac{3}{2}a^2k^2 + O(k^4),$$

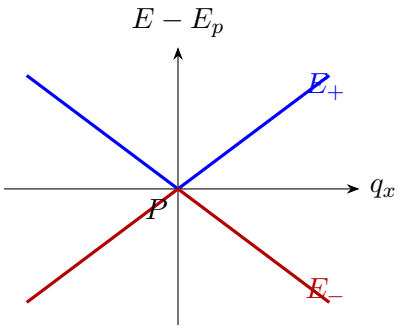
$$\sqrt{S} = 3 - \frac{a^2k^2}{4} + O(k^4), \qquad E_\pm = E_p \pm 3\gamma \mp \frac{\gamma a^2}{4}k^2 + O(k^4).$$

Hence Γ is a maximum of the upper band and a minimum of the lower band. The Hessians $\mp(\gamma a^2/2)I$ are definite.

At $P = (4\pi/(3a), 0)$, write $\boldsymbol{k} = \boldsymbol{P} + \boldsymbol{q}$. Then

$$S = \frac{3}{4}a^2(q_x^2 + q_y^2) + O(q^3), \qquad \boxed{E_\pm = E_p \pm \beta|\boldsymbol{q}| + O(q^2), \quad \beta = \frac{\sqrt{3}\gamma a}{2}.}$$

Along $q_y = 0$, the term $\beta|q_x|$ has unequal left and right derivatives. Therefore P is a **Dirac touching, not a differentiable zero-gradient critical point**. A broader terminology may call it a nonsmooth upper-band minimum and lower-band maximum, but the conical nature must be stated; a quadratic effective-mass approximation is invalid.



Section through a Dirac cone: opposite sides have different slopes.

(b) DOS around the lower-band minimum at Γ

Define the DOS per real-space area, per spin, for one local patch:

$$D(E) = \int_{\text{patch}} \frac{\mathrm{d}^2 q}{(2\pi)^2} \delta(E - E(\boldsymbol{q})).$$

Let $E_0 = E_p - 3\gamma$, $\alpha = \gamma a^2/4$, and use $E = E_0 + \alpha q^2$ in $q < \Lambda$. Then

$$\begin{aligned} D_-(E) &= \frac{1}{2\pi}\int_0^\Lambda q \, \mathrm{d}q \, \delta(E - E_0 - \alpha q^2) \\ &= \boxed{\frac{1}{4\pi\alpha}\Theta(E - E_0)\Theta(E_0 + \alpha\Lambda^2 - E)} \\ &= \frac{1}{\pi\gamma a^2}\Theta(E - E_0)\Theta(E_0 + \alpha\Lambda^2 - E). \end{aligned}$$

This is the constant step at a two-dimensional parabolic extremum. The upper-band maximum $E_1 = E_p + 3\gamma$ gives a reversed step with the same height for $E_1 - \alpha\Lambda^2 < E < E_1$. Multiply by sample area for total DOS and by two for both spins. No extra valley factor belongs to a single patch.

For comparison, one Dirac valley per spin gives

$$D_P(E) = \frac{|E - E_p|}{2\pi\beta^2} = \frac{2|E - E_p|}{3\pi\gamma^2 a^2},$$

which vanishes at the touching energy.

4. Hamiltonian with fixed nuclei

Fix nuclear coordinates $\boldsymbol{R}_A$ and charges $Z_A e$. Neglect relativity, spin–orbit coupling, and nuclear motion. Define

$$v_{\text{ext}}(\boldsymbol{r}) = -\sum_A \frac{Z_A \kappa}{|\boldsymbol{r} - \boldsymbol{R}_A|}, \qquad h(i) = -\frac{\hbar^2}{2m_e}\nabla_i^2 + v_{\text{ext}}(\boldsymbol{r}_i).$$

Then

$$\boxed{H_e = \sum_{i=1}^{N} h(i) + \sum_{i<j} \frac{\kappa}{r_{ij}}.}$$

These terms are electron kinetic energy, nuclear attraction, and electron repulsion. Total energy at the fixed nuclear configuration also includes the constant

$$E_{nn} = \sum_{A<B} \frac{Z_A Z_B \kappa}{|\boldsymbol{R}_A - \boldsymbol{R}_B|}.$$

This does not affect electronic variation. Writing the pair sum as $\sum_{i \neq j}$ requires a factor $1/2$.

5. Hartree trial state

For $x_i = (\boldsymbol{r}_i, \sigma_i)$,

$$\boxed{\Psi_H(x_1, \ldots, x_N) = \prod_{i=1}^{N} \chi_i(x_i), \qquad \int |\chi_i(x)|^2 \, \mathrm{d}x = 1.}$$

The expectation value is

$$E_H = \sum_i h_{ii} + \frac{1}{2}\sum_{i \neq j} J_{ij}, \qquad J_{ij} = \iint \frac{\kappa |\chi_i(x)|^2 |\chi_j(x')|^2}{|\boldsymbol{r} - \boldsymbol{r}'|} \, \mathrm{d}x \, \mathrm{d}x'.$$

Variation of each normalized orbital gives the average Hartree field of the other electrons. The product does not enforce antisymmetry, so it does not fully satisfy fermionic statistics and has no exchange term.

6. Hartree–Fock trial state and optimization

Antisymmetrization and energy

Take orthonormal spin orbitals:

$$\boxed{\Psi_{\text{HF}}(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det[\chi_i(x_j)]_{i,j=1}^{N}, \quad \langle \chi_i \mid \chi_j \rangle = \delta_{ij}.}$$

Exchanging two electron coordinates exchanges columns and changes the sign. Its energy is

$$E_{\text{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij} (J_{ij} - K_{ij}).$$

Since $J_{ii} = K_{ii}$, self-interaction cancels.

Constrained variation

Minimize

$$\mathcal{L} = E_{\text{HF}} - \sum_{ij} \lambda_{ij}(\langle \chi_i \mid \chi_j \rangle - \delta_{ij}).$$

Variation with respect to $\chi_i^*(x)$ gives

$$F\chi_i = \sum_j \lambda_{ij}\chi_j, \qquad F = h + \sum_{j\in\text{occ}} (J_j - K_j),$$

$$(J_j f)(x) = \left[\int \frac{\kappa|\chi_j(x')|^2}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}x'\right] f(x),$$
$$(K_j f)(x) = \chi_j(x) \int \frac{\kappa\chi_j^*(x')f(x')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}x'.$$

J_j is local; K_j is nonlocal. Diagonalize the Hermitian multiplier matrix by a unitary rotation of occupied orbitals, changing the determinant only by an overall phase. The canonical equations are

$$\boxed{F\chi_i = \epsilon_i\chi_i.}$$

Self-consistency and limitations

Guess orbitals, build the density and Fock operator, solve its eigenproblem, occupy N orbitals, and update until density and energy converge. Compare distinct spin or symmetry solutions; convergence alone does not prove the global minimum. The optimized determinant is a variational approximation in a restricted space, not generally an exact correlated state. Its total energy is

$$E_{\text{HF}} = \sum_i \epsilon_i - \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}),$$

not the uncorrected eigenvalue sum.

7. Slater–Jastrow trial state

For a fluid electron gas in periodic volume Ω, use momentum-state determinants times a positive pair-correlation factor:

$$\boxed{\Psi_T(\boldsymbol{R}) = D_\uparrow D_\downarrow \exp\left[-\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij})\right], \qquad \boldsymbol{R} = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_N).}$$

For example $D_\sigma = \det[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_\alpha\cdot\boldsymbol{r}_j}/\sqrt{\Omega}]$ fills the chosen Fermi sea for each spin population. This is a fixed-spin coordinate component; the complete spin wavefunction remains antisymmetric. A fully polarized state has one spin determinant.

The determinants supply antisymmetry and nodes. The symmetric positive Jastrow factor preserves the nodes while changing pair probabilities and lowering correlation energy. Its short-range form can satisfy Coulomb cusp conditions; its long-range form can incorporate screening and collective fluctuations. Optimize its parameters variationally. A trial crystal phase can instead use localized orbitals around lattice sites.

Sampling $p(\boldsymbol{R}) \propto |\Psi_T(\boldsymbol{R})|^2$ gives the local-energy average:

$$E_T = \frac{\int |\Psi_T|^2 E_L \, \mathrm{d}\boldsymbol{R}}{\int |\Psi_T|^2 \, \mathrm{d}\boldsymbol{R}}, \quad E_L = \frac{H\Psi_T}{\Psi_T}, \quad E_T \geq E_0.$$

Projection Monte Carlo further filters higher energies with imaginary-time evolution $\mathrm{e}^{-\tau(H-E_T)}$. Holding trial nodes fixed yields a fixed-node approximation, not an error-free exact result. The Ceperley–Alder stochastic projection study is more than a single variational sampling calculation.

Primary reference used for checks

D. M. Ceperley and B. J. Alder, Ground State of the Electron Gas by a Stochastic Method, Phys. Rev. Lett. 45, 566 (1980). An author-hosted full text is available. The lattice geometry, band expansions, DOS factors, and variational steps were derived independently.

Chapter 3

Solutions: Midterm Exam, April 30, 2015

These solutions are derived from the stated assumptions. The density of states $D(E)$ counts states per unit energy in the entire volume V. The factor g_s denotes any spin degeneracy included separately; the free-electron gas uses $g_s = 2$ without double counting.

1. Lattice translations in $BaTiO_3$

For the cubic model specified in the question, the Bravais translations are $\boldsymbol{R} = a(n_x, n_y, n_z)$ with $n_i \in \mathbb{Z}$. A five-atom basis is

Atom	$\boldsymbol{\tau}/a$
Ba	$(0, 0, 0)$
Ti	$(1/2, 1/2, 1/2)$
O_1	$(1/2, 1/2, 0)$
O_2	$(1/2, 0, 1/2)$
O_3	$(0, 1/2, 1/2)$

Actual atomic positions are $\boldsymbol{R} + \boldsymbol{\tau}_\nu$. A displacement is a Bravais translation precisely when it belongs to $a\mathbb{Z}^3$.

(a) **True.** There is one Ti basis position in each primitive cell, so subtraction cancels the basis position:

$$(\boldsymbol{R}_m + \boldsymbol{\tau}_{\mathrm{Ti}}) - (\boldsymbol{R}_n + \boldsymbol{\tau}_{\mathrm{Ti}}) = \boldsymbol{R}_m - \boldsymbol{R}_n.$$

This is always a lattice translation.

(b) **True.** A Ba-to-Ti displacement has the form $a(n_x + 1/2, n_y + 1/2, n_z + 1/2)$. Its half-integer components exclude it from $a\mathbb{Z}^3$.

(c) **False.** Two O atoms with the same basis index are related by a translation, but the assertion concerns any two O atoms. Within one cell,

$$\boldsymbol{\tau}_{\mathrm{O}_2} - \boldsymbol{\tau}_{\mathrm{O}_1} = \frac{a}{2}(0, -1, 1),$$

which is not a simple-cubic lattice translation. Having the same chemical species does not imply having the same basis index.

2. The fcc lattice, its reciprocal lattice, and lattice planes

(a) Primitive-cell volumes

The given vectors are primitive vectors of a face-centered cubic (fcc) lattice. Their scalar triple product is

$$\Omega = |\boldsymbol{t}_1 \cdot (\boldsymbol{t}_2 \times \boldsymbol{t}_3)| = \left(\frac{a}{2}\right)^3 \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \frac{a^3}{4}.$$

A Wigner–Seitz cell is also a primitive cell. The reciprocal primitive-cell volume, equal to the first Brillouin-zone volume, is $(2\pi)^3/\Omega$. Thus

$$\boxed{V_{\text{WS}} = \frac{a^3}{4}}, \qquad \boxed{V_{\text{BZ}} = \frac{32\pi^3}{a^3}}.$$

(b) Reciprocal primitive vectors

Using $\boldsymbol{g}_1 = 2\pi(\boldsymbol{t}_2 \times \boldsymbol{t}_3)/\Omega$ and cyclic permutations gives

$$\boxed{\boldsymbol{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \boldsymbol{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \boldsymbol{g}_3 = \frac{2\pi}{a}(1,1,-1)}.$$

For example, $\boldsymbol{t}_1 \cdot \boldsymbol{g}_1 = 2\pi$ and $\boldsymbol{t}_1 \cdot \boldsymbol{g}_2 = \boldsymbol{t}_1 \cdot \boldsymbol{g}_3 = 0$. The other rows give the same reciprocal orthogonality relation.

(c) Spacing of successive occupied lattice planes

The unit normal is $\hat{\boldsymbol{n}} = (0,1,1)/\sqrt{2}$. Project an arbitrary lattice point $\boldsymbol{R} = n_1\boldsymbol{t}_1 + n_2\boldsymbol{t}_2 + n_3\boldsymbol{t}_3$ onto this direction:

$$\hat{\boldsymbol{n}} \cdot \boldsymbol{R} = \frac{a}{2\sqrt{2}}(2n_1 + n_2 + n_3).$$

The integer in parentheses can take every integer value. Successive planes containing fcc lattice points therefore lie at $\hat{\boldsymbol{n}} \cdot \boldsymbol{R} = ja/(2\sqrt{2})$, giving

$$\boxed{d = \frac{a}{2\sqrt{2}}}.$$

Equivalently, the shortest nonzero reciprocal vector parallel to the normal is $\boldsymbol{G} = 2\boldsymbol{g}_1 + \boldsymbol{g}_2 + \boldsymbol{g}_3 = (4\pi/a)(0,1,1)$, and $d = 2\pi/|\boldsymbol{G}|$.

> Here "successive lattice planes" means successive occupied fcc layers. The formal conventional-cubic (011) spacing is $a/\sqrt{2}$, but fcc centering supplies an additional occupied layer halfway between those planes. The occupied-layer spacing requested under this interpretation is consequently half as large.

(d) Areal density

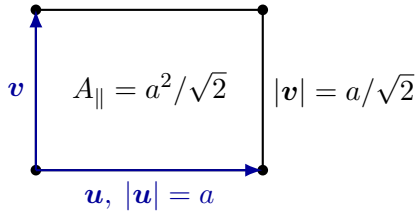
Two perpendicular in-plane translations are

$$\boldsymbol{u} = a(1,0,0) = -\boldsymbol{t}_1 + \boldsymbol{t}_2 + \boldsymbol{t}_3, \qquad \boldsymbol{v} = \frac{a}{2}(0,1,-1) = \boldsymbol{t}_3 - \boldsymbol{t}_2.$$

They generate the two-dimensional lattice on this plane. Its primitive area is $A_\parallel = |\boldsymbol{u} \times \boldsymbol{v}| = a^2/\sqrt{2}$, with one lattice point per primitive parallelogram. Hence

$$\boxed{\sigma = \frac{1}{A_\parallel} = \frac{\sqrt{2}}{a^2}}.$$

The volume check is $A_\parallel d = \Omega$, equivalently $\sigma = d/\Omega$.



Four corners contribute $1/4$ point each.

(e) Testing the proposed vectors

Every reciprocal vector must satisfy $\boldsymbol{t}_i \cdot \boldsymbol{G} \in 2\pi\mathbb{Z}$ for all primitive translations. However,

$$\boldsymbol{t}_1 \cdot \boldsymbol{G}_1 = \frac{a}{2}(0,1,1) \cdot \frac{\pi}{a}(0,1,1) = \pi.$$

Thus $\boldsymbol{G}_1$ is not even a reciprocal-lattice vector. The proposed vectors **do not** generate the required reciprocal lattice by integer linear combinations. Linear independence over the real numbers only guarantees that three vectors span three-dimensional vector space; it does not establish equality of discrete lattices.

3. Atomic density of diamond

Diamond consists of an fcc Bravais lattice with a two-atom basis. A conventional cubic cell contains $8(1/8) + 6(1/2) = 4$ fcc lattice points. With two atoms per point,

$$\boxed{n_{\text{atom}} = \frac{4 \times 2}{a^3} = \frac{8}{a^3}}.$$

The equivalent primitive-cell count is two atoms divided by $a^3/4$.

4. Definitions

(a) **Three-dimensional hexagonal lattice.** One choice of primitive vectors is

$$\boldsymbol{a}_1 = a(1,0,0), \quad \boldsymbol{a}_2 = a(-1/2, \sqrt{3}/2, 0), \quad \boldsymbol{a}_3 = c(0,0,1).$$

The first two vectors have equal lengths and enclose $120°$; the third is perpendicular to their plane. Their integer combinations define the hexagonal Bravais lattice, of primitive volume $\sqrt{3}a^2c/2$. Hexagonal close packing (hcp) adds a two-atom basis to this lattice; the lattice and its basis are separate concepts.

(b) **Bloch theorem.** If a single-electron Hamiltonian is invariant under every lattice translation, its energy eigenstates can be chosen as

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}).$$

Equivalently, $\psi_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \psi_{n\boldsymbol{k}}(\boldsymbol{r})$. The wave vector labels crystal momentum and can be restricted to the first Brillouin zone modulo a reciprocal vector.

(c) **Critical point.** A point where $\nabla_{\boldsymbol{k}}\varepsilon_n = 0$. For a nondegenerate Hessian in three dimensions, the type M_j counts its negative eigenvalues: M_0 is a minimum, M_1 and M_2 are saddles with one and two negative directions, and M_3 is a maximum.

(d) **Fermi energy.** The highest occupied single-particle energy after filling the zero-temperature ground state at fixed particle number. For the free-electron gas here it equals $\mu(0) = \varepsilon_F$. Its energy reference must be specified; it is distinct from the work function required to remove an electron to vacuum.

5. Density of states near a three-dimensional M_1 saddle

(a) Local dispersion

Set $\boldsymbol{q} = \boldsymbol{k} - \boldsymbol{k}_1$ and diagonalize the Hessian. The gradient vanishes at the critical point, so no linear term remains. With one negative-curvature direction,

$$\boxed{\varepsilon(\boldsymbol{k}) = \varepsilon_1 + Aq_x^2 + Bq_y^2 - Cq_z^2 + O(q^3)}, \qquad A, B, C > 0.$$

The coefficients A, B, C already incorporate the factor $1/2$ in the Taylor expansion.

(b) Deriving the nonanalytic contribution with a local cutoff

Define $X = \sqrt{A}q_x$, $Y = \sqrt{B}q_y$, $Z = \sqrt{C}q_z$, and $\Delta = E - \varepsilon_1$. Then

$$D_{\text{loc}}(E) = \frac{g_s V}{(2\pi)^3\sqrt{ABC}} \int \mathrm{d}X\, \mathrm{d}Y\, \mathrm{d}Z\, \delta(\Delta - X^2 - Y^2 + Z^2).$$

The quadratic saddle cannot be extended to infinity: its constant-energy surfaces are unbounded. Restrict it to a neighborhood, for example $|Z| < \Lambda$, with a transverse cutoff large enough to contain the relevant energy shells in the range considered. Polar integration over X, Y gives

$$\int_0^\infty 2\pi\rho\, \mathrm{d}\rho\, \delta(\Delta + Z^2 - \rho^2) = \pi\Theta(\Delta + Z^2).$$

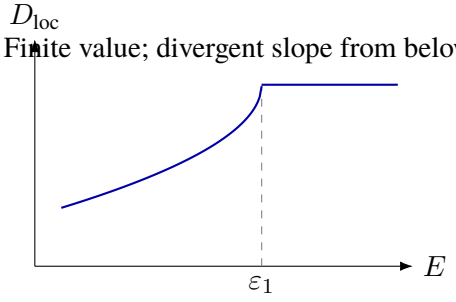
For $\Delta > 0$, every Z in $(-\Lambda, \Lambda)$ contributes. For $-\Lambda^2 < \Delta < 0$, the interval $|Z| < \sqrt{-\Delta}$ has no real transverse radius and must be removed. Therefore

$$D_{\text{loc}}(E) = \frac{g_s V}{4\pi^2\sqrt{ABC}} \left[\Lambda - \sqrt{\varepsilon_1 - E}\,\Theta(\varepsilon_1 - E)\right].$$

Other cutoff choices and other parts of the band change the regular background, while the leading local nonanalytic term is

$$\boxed{D(E) = D_{\text{reg}}(E) - \frac{g_s V}{4\pi^2\sqrt{ABC}}\sqrt{\varepsilon_1 - E}\,\Theta(\varepsilon_1 - E)}.$$

This expression gives the leading behavior near the critical energy; higher-order terms in the dispersion can add subleading corrections. The density of states remains finite at $E = \varepsilon_1$, but its slope diverges on the lower-energy side. Local curvature alone cannot fix the absolute background D_{reg} of the whole band.



6. Free-electron ground-state quantities in terms of r_s

Let $a_0 = 4\pi\varepsilon_0\hbar^2/(m_e e^2)$ be the Bohr radius. By definition, a sphere of radius $r_s a_0$ contains the mean volume per electron:

$$\frac{1}{n} = \frac{4\pi}{3}(r_s a_0)^3 \quad \Longrightarrow \quad \boxed{n = \frac{3}{4\pi r_s^3 a_0^3}}.$$

Counting both spins inside the Fermi sphere gives

$$N = 2\frac{V}{(2\pi)^3}\frac{4\pi}{3}k_F^3, \qquad n = \frac{k_F^3}{3\pi^2}.$$

Consequently,

$$\boxed{k_F = \frac{(9\pi/4)^{1/3}}{r_s a_0}}, \qquad \boxed{\varepsilon_F = \frac{\hbar^2}{2m_e a_0^2}\left(\frac{9\pi}{4}\right)^{2/3}\frac{1}{r_s^2}},$$

$$\boxed{T_F = \frac{\varepsilon_F}{k_\mathrm{B}}}.$$

The mean energy averages all occupied states, rather than only states on the Fermi surface:

$$\overline{\varepsilon} = \frac{\int_0^{k_F}(\hbar^2k^2/2m_e)k^2\,\mathrm{d}k}{\int_0^{k_F}k^2\,\mathrm{d}k} = \boxed{\frac{3}{5}\varepsilon_F}.$$

7. Low-temperature heat capacity and the printed coefficient

For $T \ll T_F$, states well below the Fermi energy are almost completely filled. Pauli exclusion prevents arbitrary changes of their occupations. Thermal excitations predominantly involve an energy window of width of order $k_\mathrm{B}T$ near ε_F. Since $D(\varepsilon_F) = 3N/(2\varepsilon_F)$ in three dimensions, the number of thermally active states scales as

$$N_\mathrm{active} \sim D(\varepsilon_F)k_\mathrm{B}T = \frac{3}{2}N\frac{T}{T_F}.$$

Assigning a heat-capacity scale k_B to each active state gives the estimate printed in the question: a suppression by T/T_F relative to the classical value.

The printed factor $3/2$ is not the exact low-temperature coefficient. The Sommerfeld expansion at fixed N, V gives

$$C_V = \frac{\pi^2}{3}k_\mathrm{B}^2TD(\varepsilon_F) = \boxed{\frac{\pi^2}{2}Nk_\mathrm{B}\frac{T}{T_F}} + O\big(Nk_\mathrm{B}(T/T_F)^3\big)\,.$$

Thus the physical argument establishes linear temperature dependence and a restricted thermally active population. It must not be interpreted as an exact derivation of the numerical coefficient $3/2$.

8. The exponential tail above the vacuum level

The Fermi–Dirac distribution is

$$f(E) = \frac{1}{\mathrm{e}^{(E-\mu)/(k_\mathrm{B}T)}+1}.$$

Define the work function $\Phi = E_\mathrm{vac} - \mu > 0$. When $\Phi \gg k_\mathrm{B}T$, as in the usual low-temperature metallic regime, all $E \geq E_\mathrm{vac}$ satisfy $x = (E-\mu)/(k_\mathrm{B}T) \geq \Phi/(k_\mathrm{B}T) \gg 1$. It follows that

$$f(E) = \frac{\mathrm{e}^{-x}}{1+\mathrm{e}^{-x}} \simeq \boxed{\exp[-(E-\mu)/(k_\mathrm{B}T)]}.$$

The relative error of $f_{\text{exp}} = \mathrm{e}^{-x}$ is exactly

$$\frac{f_{\text{exp}} - f}{f} = \mathrm{e}^{-x} \leq \mathrm{e}^{-\Phi/(k_{\mathrm{B}}T)}.$$

The approximation is controlled by the energy separation from the chemical potential compared with thermal energy, not by the word "vacuum" alone.

References and checks

The counting arguments and integrals are derived above. Primary teaching references for comparison are David Tong, Statistical Physics, Chapter 3 (state counting and the Sommerfeld expansion), and M. S. Dresselhaus, MIT Solid State Physics Part II, Chapter 4 (critical-point classification and the three-dimensional saddle-point singularity).

Chapter 4

Solutions: Final Exam, June 25, 2015

The electronic and vibrational conventions are stated explicitly below. In particular, a band is counted before separately resolving spin unless otherwise specified, and phonon polarization vectors are mass-weighted as in the hint.

1. Bloch sums and their role in tight binding

Let $\phi_{\nu a}(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\nu)$ be a localized orbital of type a on basis atom ν in the cell at $\boldsymbol{R}$. For N cells with periodic boundary conditions, define

$$\boxed{\Phi_{\nu a,\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \phi_{\nu a}(\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\tau}_\nu)}.$$

The factor $N^{-1/2}$ is conventional; if orbitals on different sites overlap, the overlap matrix still determines the final normalization.

For any Bravais translation $\boldsymbol{T}$, reindex the sum by $\boldsymbol{R}' = \boldsymbol{R} - \boldsymbol{T}$:

$$\begin{aligned}\Phi_{\nu a,\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{T}) &= \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}'} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}'+\boldsymbol{T})} \phi_{\nu a}(\boldsymbol{r}-\boldsymbol{R}'-\boldsymbol{\tau}_\nu) \\ &= \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{T}} \Phi_{\nu a,\boldsymbol{k}}(\boldsymbol{r}).\end{aligned}$$

The sum therefore has precisely the Bloch translation property.

In tight binding, these symmetry-adapted localized-orbital sums form the basis at each wave vector:

$$\psi_{n\boldsymbol{k}} = \sum_{\nu a} c_{\nu a}^{(n)}(\boldsymbol{k}) \Phi_{\nu a,\boldsymbol{k}}.$$

Projecting the Schrödinger equation onto this basis gives

$$\boxed{H(\boldsymbol{k})c_n = E_n(\boldsymbol{k}) S(\boldsymbol{k}) c_n},$$

where $H_{ij} = \langle\Phi_i|\hat{H}|\Phi_j\rangle$ and $S_{ij} = \langle\Phi_i|\Phi_j\rangle$. In an orthonormal model, $S = I$. Translational symmetry has reduced an extensive real-space problem to a finite matrix at each $\boldsymbol{k}$; hopping between localized orbitals determines the matrix entries and their wave-vector-dependent phases.

2. Five orbitals per Si atom and the band count

For electrical and bonding properties, retain the valence $3s$ and three $3p$ orbitals. A useful fifth orbital is an excited effective s-like orbital s^*, giving the conventional

$$\boxed{\{3s, 3p_x, 3p_y, 3p_z, s^*\} \quad (sp^3s^*)}$$

basis. The s^* orbital improves the description of conduction bands; in an empirical model it is an effective excited orbital, not a claim that the isolated atom already occupies a fifth valence orbital. The filled $1s$, $2s$, and $2p$ shells are treated as core states for this low-energy model. The original model is described by Vogl, Hjalmarson, and Dow (1983).

Silicon has two atoms per diamond primitive cell. Five orbital functions per atom therefore yield

$$\boxed{2 \times 5 = 10 \text{ bands before explicit spin resolution}}.$$

Each atom supplies four valence electrons, so each primitive cell supplies eight. A filled band accommodates two electrons per primitive cell when both spins are included. Hence

$$\boxed{\frac{2 \times 4}{2} = 4 \text{ fully occupied valence bands}}.$$

The other six bands of this model are unoccupied in the undoped ground state. If spin is represented explicitly, the matrix has dimension twenty and eight spin-resolved bands are occupied. These conventions are consistent with the nextnano implementation of the sp^3s^* model.

3. Why a direct all-electron plane-wave expansion is expensive

At fixed $\boldsymbol{k}$, a Bloch wave can be expanded as

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{G}} c_{n\boldsymbol{k}}(\boldsymbol{G}) \mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}.$$

Near a nucleus, the all-electron potential varies strongly and the wavefunctions contain short spatial scales. Valence functions must also remain orthogonal to localized core states, producing nodes and rapid core-region variation. Resolving a length scale ℓ requires components with $|\boldsymbol{k}+\boldsymbol{G}|$ of order $1/\ell$. A kinetic-energy cutoff

$$\frac{\hbar^2|\boldsymbol{k}+\boldsymbol{G}|^2}{2m_e} \leq E_{\text{cut}}$$

then requires a very large E_{cut} and many plane waves, even if only valence energies are sought. The basis is valid, but direct all-electron convergence can be inefficient.

One remedy is to replace the core-region problem by a pseudopotential with smooth pseudo-valence functions. Alternatively, augmented-plane-wave or projector-augmented-wave approaches retain or reconstruct atomic core-region structure while using smoother representations elsewhere. Orthogonalized-plane-wave methods also account explicitly for the core orthogonality constraint. In each case, the objective is to represent the sharp atomic behavior without forcing a uniform plane-wave basis to resolve every core length scale directly.

4. Pseudopotentials and smooth pseudo-wavefunctions

A pseudopotential is an effective electron–ion operator replacing the explicit nuclear attraction and core-electron effects in a valence calculation. It is constructed to reproduce relevant valence eigenvalues and

scattering behavior outside a chosen core region. It can be angular-momentum dependent and nonlocal; it need not be a single scalar function of position.

A pseudo-wavefunction follows the required valence behavior outside the core but replaces the rapid all-electron oscillations inside it by a smoother function. Its large-$|\boldsymbol{G}|$ Fourier coefficients are therefore much smaller. A substantially lower plane-wave cutoff can represent it to a chosen accuracy.

This is an approximation with a chosen core–valence partition and transferability range. Smoothness does not itself guarantee correct bonding in every environment, and the cutoff must still be converged. Practical construction and validation requirements are discussed in Quantum ESPRESSO, Notes on pseudopotential generation.

5. The adiabatic energy surface

Collect all nuclear positions into $\boldsymbol{R}$ and the electronic coordinates into $\boldsymbol{r}$. For fixed nuclei, solve the electronic eigenproblem

$$\hat{H}_e(\boldsymbol{R})\Psi_n(\boldsymbol{r};\boldsymbol{R}) = \mathcal{E}_n(\boldsymbol{R})\Psi_n(\boldsymbol{r};\boldsymbol{R}).$$

The adiabatic potential-energy surface for nuclear motion is

$$\boxed{E_n(\boldsymbol{R}) = \mathcal{E}_n(\boldsymbol{R}) + V_{NN}(\boldsymbol{R})},$$

where V_{NN} is the nucleus–nucleus repulsion. If that term has already been included in $\hat{H}_e$, it must not be added a second time.

The electronic eigenproblem is solved parametrically as the nuclei move. In the Born–Oppenheimer approximation, nuclei move on a selected surface, neglecting transitions to other electronic surfaces and the associated nonadiabatic couplings. Nuclear kinetic energy is not part of this potential-energy surface. Its gradient determines the forces, and its Hessian determines harmonic restoring forces about equilibrium.

6. Obtaining lattice normal modes from the ground surface

Equilibrium and harmonic force constants

First locate a stationary nuclear configuration $\boldsymbol{R}^0$ with $\partial E_0/\partial R_{i\alpha} = 0$. Let $u_{i\alpha} = R_{i\alpha} - R^0_{i\alpha}$, where i labels atoms and α labels Cartesian components. Expand

$$E_0(\boldsymbol{R}^0 + \boldsymbol{u}) = E_0(\boldsymbol{R}^0) + \frac{1}{2}\sum_{i\alpha,j\beta}\Phi_{i\alpha,j\beta}u_{i\alpha}u_{j\beta} + O(u^3), \qquad \Phi_{i\alpha,j\beta} = \left.\frac{\partial^2 E_0}{\partial R_{i\alpha}\partial R_{j\beta}}\right|_{\boldsymbol{R}^0}.$$

The linear term vanishes at equilibrium. The Hessian is the harmonic force-constant matrix, obtained analytically or by converged finite differences of energies or forces.

Mass weighting and the eigenproblem

Newton's equations in the harmonic approximation are

$$M_i\ddot{u}_{i\alpha} = -\sum_{j\beta}\Phi_{i\alpha,j\beta}u_{j\beta}.$$

Substitute $u_{i\alpha}(t) = A_{i\alpha}\mathrm{e}^{-\mathrm{i}\omega t}/\sqrt{M_i}$. This gives

$$\sum_{j\beta}\mathcal{D}_{i\alpha,j\beta}A_{j\beta} = \omega^2 A_{i\alpha}, \qquad \boxed{\mathcal{D}_{i\alpha,j\beta} = \frac{\Phi_{i\alpha,j\beta}}{\sqrt{M_iM_j}}}.$$

Diagonalizing the real-space mass-weighted Hessian yields the squared normal-mode frequencies and orthonormal polarization vectors. The physical displacements are obtained by dividing each atomic block of an eigenvector by $\sqrt{M_i}$.

Periodic crystal and stability

For a periodic crystal with r atoms per primitive cell, Fourier transformation block-diagonalizes the problem into a $3r \times 3r$ dynamical matrix at each allowed $\boldsymbol{q}$. The next answer gives its entries. The branches $\omega_p(\boldsymbol{q})$ and their eigenvectors are the lattice normal modes.

Translational invariance imposes the acoustic sum rule

$$\sum_j \Phi_{i\alpha,j\beta} = 0$$

for every i, α, β. Thus three rigid translations have zero frequency at $\boldsymbol{q} = 0$. A mechanically stable harmonic equilibrium has no negative ω^2; negative eigenvalues describe an instability rather than ordinary oscillations. A nondegenerate, well-separated *electronic* ground surface justifies the adiabatic treatment but does not by itself prove that a chosen nuclear configuration is a local minimum.

7. Dynamical-matrix elements and phase convention

To remove a notational ambiguity, use the symbol D exactly as in the question for the force constants, and $\mathcal{D}$ for the mass-weighted dynamical matrix. Adopt

$$D_{\nu\alpha,\nu'\alpha'}(\boldsymbol{t}_n) = \left.\frac{\partial^2 E_0}{\partial u_{0\nu\alpha}\,\partial u_{n\nu'\alpha'}}\right|_0 .$$

Choose the displacement convention

$$u_{n\nu\alpha}(t) = \frac{A_{\nu\alpha}(\boldsymbol{q})}{\sqrt{M_\nu}} \mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{t}_n - \mathrm{i}\omega t}.$$

Inserting this into the equation of motion for atom $(0,\nu)$ gives

$$\boxed{\mathcal{D}_{\nu\alpha,\nu'\alpha'}(\boldsymbol{q}) = \frac{1}{\sqrt{M_\nu M_{\nu'}}} \sum_n D_{\nu\alpha,\nu'\alpha'}(\boldsymbol{t}_n)\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{t}_n}},$$

and the eigenproblem

$$\sum_{\nu'\alpha'} \mathcal{D}_{\nu\alpha,\nu'\alpha'}(\boldsymbol{q}) A_{\nu'\alpha'} = \omega^2 A_{\nu\alpha}.$$

Hessian symmetry and translation invariance give $D_{\nu\alpha,\nu'\alpha'}(\boldsymbol{t}_n) = D_{\nu'\alpha',\nu\alpha}(-\boldsymbol{t}_n)$, so $\mathcal{D}(\boldsymbol{q})$ is Hermitian.

If the plane-wave factor is instead defined using the full equilibrium position $\boldsymbol{t}_n + \boldsymbol{\tau}_\nu$, the exponential in the boxed matrix becomes $\mathrm{e}^{\mathrm{i}\boldsymbol{q}\cdot(\boldsymbol{t}_n+\boldsymbol{\tau}_{\nu'}-\boldsymbol{\tau}_\nu)}$. The two matrices are related by a diagonal unitary change of basis and have identical frequencies. Stating the convention prevents an apparent phase disagreement from becoming a physical one.

8. Counting $BaTiO_3$ vibrational degrees of freedom

There are $r = 5$ atoms per cell and three displacement coordinates per atom.

(a) At each $\boldsymbol{q}$, the Fourier-space dynamical matrix acts on $3r = 15$ coordinates, so its dimension is

$$\boxed{15 \times 15}.$$

If "dynamical matrix" instead refers to the complete real-space system of N cells, its dimension is $15N \times 15N$. The standard band/branch formulation uses the first convention.

(b) There are N allowed wave vectors and fifteen modes at each one:

$$\boxed{15N \text{ normal modes}}.$$

The real-displacement condition relates complex amplitudes at $\boldsymbol{q}$ and $-\boldsymbol{q}$; it does not halve the total number of real mechanical degrees of freedom.

(c) The three rigid translations evolve into three acoustic branches:

$$\boxed{3 \text{ acoustic branches}}.$$

(d) The remaining branches are optical:

$$\boxed{3r - 3 = 12 \text{ optical branches}}.$$

Each branch contains N normal modes, so the corresponding mode counts are $3N$ acoustic and $12N$ optical.

9. Long-wavelength optical polarization vectors

The hint uses *mass-weighted* vectors $\boldsymbol{A}_\nu$, not the actual atomic displacements. At $\boldsymbol{q} = 0$, a rigid translation in direction $\hat{\boldsymbol{e}}_\alpha$ has a mass-weighted eigenvector proportional to

$$\left(\sqrt{M_1}\hat{\boldsymbol{e}}_\alpha, \sqrt{M_2}\hat{\boldsymbol{e}}_\alpha\right).$$

A finite-frequency optical eigenvector of the Hermitian dynamical matrix is orthogonal to each of the three zero-frequency translational vectors. Component by component this implies

$$\sqrt{M_1}\boldsymbol{A}_1 + \sqrt{M_2}\boldsymbol{A}_2 = 0, \qquad \boldsymbol{A}_2 = -\sqrt{\frac{M_1}{M_2}}\boldsymbol{A}_1.$$

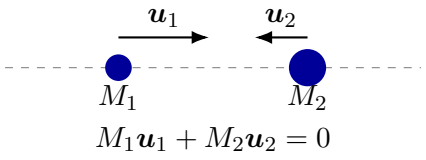
Therefore

$$\boldsymbol{A}_1^* \cdot \boldsymbol{A}_2 = -\sqrt{\frac{M_1}{M_2}}\left|\boldsymbol{A}_1\right|^2, \qquad \boxed{\frac{\left|\boldsymbol{A}_1\right|^2}{\boldsymbol{A}_1^* \cdot \boldsymbol{A}_2} = -\sqrt{\frac{M_2}{M_1}}}.$$

The result is unchanged by a common complex phase of the mode. In terms of the *physical* displacement amplitudes $\boldsymbol{u}_\nu = \boldsymbol{A}_\nu/\sqrt{M_\nu}$, the corresponding relation is

$$\boldsymbol{u}_2 = -\frac{M_1}{M_2}\boldsymbol{u}_1, \qquad M_1\boldsymbol{u}_1 + M_2\boldsymbol{u}_2 = 0.$$

The unit-cell center of mass is stationary in the optical limit. The square-root mass ratio in the requested answer arises specifically from the mass-weighted convention in the hint.



$$M_1\boldsymbol{u}_1 + M_2\boldsymbol{u}_2 = 0$$

The schematic shows opposite motion; arrow lengths are illustrative. At an exactly soft optical instability, the optical subspace must be selected orthogonally to translations before taking the zero-frequency limit. The standard finite-frequency optical limit is assumed in the question.

References and checks

The preceding formulas follow from the displayed translation, state-counting, and Hessian arguments. The primary sources used to check conventions are:

- P. Vogl, H. P. Hjalmarson, and J. D. Dow, A Semi-empirical tight-binding theory of the electronic structure of semiconductors (1983): the sp^3s^* basis.
- nextnano, Empirical tight-binding sp^3s^* band structure: ten-orbital and twenty-spin-orbital matrix conventions.
- Quantum ESPRESSO, Notes on pseudopotential generation: effective-core construction and validation.
- Quantum ESPRESSO, dynmat.x documentation: orthogonal dynamical-matrix eigenvectors versus physical displacements divided by square roots of masses.
- David Tong, Solid State Physics, Chapter 4: lattice vibrations and acoustic/optical branches.

Chapter 5

Solutions: Midterm, 29 April 2016

All twelve printed questions are available. The solutions below distinguish exact statements from approximations and make the assumptions behind the true/false questions explicit. In Questions 7–10, ψ always denotes the physical crystal state and φ its auxiliary pseudo wavefunction; reversing these symbols reverses the projection relation.

1. True or false

(a) A self-consistent HF determinant is an exact interacting eigenstate: false

Hartree–Fock minimizes the expectation value of the interacting Hamiltonian within the set of single Slater determinants:

$$E_{\mathrm{HF}} = \min_{\Phi \text{ determinant}} \frac{\langle \Phi | H | \Phi \rangle}{\langle \Phi \mid \Phi \rangle} \geq E_0.$$

Stationarity within this restricted set does not imply $H\Phi = E\Phi$ in the full many-electron Hilbert space. The two-electron interaction generally couples the HF determinant to doubly excited determinants. Brillouin's theorem removes its direct coupling to singly excited determinants at a stationary HF solution, not to all other states. Exceptional systems may have an exact single-determinant ground state; the assertion for general interacting electrons is false.

(b) Increasing density favors complete spin polarization: false

Let $n = N/\Omega$, $\zeta = (n_\uparrow - n_\downarrow)/n$, and $k_F = (3\pi^2 n)^{1/3}$ for the unpolarized gas. The noninteracting kinetic energy and HF exchange energy per electron are

$$t(n,\zeta) = \frac{3\hbar^2 k_F^2}{10 m_e} \frac{(1+\zeta)^{5/3} + (1-\zeta)^{5/3}}{2},$$
$$e_x(n,\zeta) = -\frac{3\kappa k_F}{4\pi} \frac{(1+\zeta)^{4/3} + (1-\zeta)^{4/3}}{2}, \quad \kappa = \frac{e^2}{4\pi\epsilon_0}.$$

The neutralizing background cancels the uniform direct electrostatic contribution. Comparing $\zeta = 1$ with $\zeta = 0$,

$$\frac{t(n,1)}{t(n,0)} = 2^{2/3} > 1, \qquad \frac{e_x(n,1)}{e_x(n,0)} = 2^{1/3} > 1.$$

Polarization lowers the negative exchange energy but raises the kinetic energy because all electrons must fill one spin species to a larger Fermi momentum. The penalty scales as $n^{2/3}$ whereas the exchange gain scales as $n^{1/3}$; thus sufficiently high density favors the unpolarized state. This scaling argument does not

determine the exact intermediate-density phase boundary, where correlation matters, nor the eventual low-density Wigner crystal.

(c) Odd electron count excludes every electrical insulator: false as stated

For a nonmagnetic independent-electron crystal with two spin states per Bloch band and the specified primitive translation symmetry, one completely filled band contains two electrons per primitive cell. Odd filling therefore forces an incompletely filled band and excludes an ordinary spin-degenerate band insulator.

The question, however, says "electrical insulator" without these assumptions. A half-filled one-orbital Hubbard system has one electron per primitive cell and can be a Mott insulator: in the atomic limit $t = 0$, moving an electron to a neighboring occupied site creates a doublon and a hole and costs $U > 0$. Its charge excitations are gapped despite odd filling. Magnetic ordering or a larger ordered unit cell can also invalidate the simple counting argument. Thus the restricted band-theory statement is true, but the unrestricted printed statement is false.

2. Fixed-nuclei many-electron Hamiltonian

For fixed nuclear positions $\boldsymbol{R}_A$ and charges $Z_A e$, define

$$v_{\text{ext}}(\boldsymbol{r}) = -\sum_A \frac{Z_A \kappa}{|\boldsymbol{r} - \boldsymbol{R}_A|}, \qquad h(i) = -\frac{\hbar^2}{2m_e}\nabla_i^2 + v_{\text{ext}}(\boldsymbol{r}_i).$$

The nonrelativistic electronic Hamiltonian is

$$\boxed{H_e = \sum_{i=1}^{N} h(i) + \sum_{i<j} \frac{\kappa}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}.}$$

These are the electron kinetic energy, electron–nucleus attraction, and electron–electron repulsion. The restriction $i < j$ counts each pair once. If the requested energy includes the nuclear electrostatic energy, add

$$E_{nn} = \sum_{A<B} \frac{Z_A Z_B \kappa}{|\boldsymbol{R}_A - \boldsymbol{R}_B|}.$$

This is constant for the fixed configuration and does not affect the electronic variational equations. Nuclear kinetic energy is absent under the fixed-nuclei assumption.

3. Hartree and HF trial states

Write a space–spin coordinate as $x = (\boldsymbol{r}, \sigma)$, with $\int \mathrm{d}x$ including the sum over spin. The Hartree state is a product of normalized one-electron functions:

$$\boxed{\Psi_H(x_1, \ldots, x_N) = \prod_{i=1}^{N} \chi_i(x_i), \qquad \int |\chi_i(x)|^2 \, \mathrm{d}x = 1.}$$

It treats other electrons through an average field but does not enforce fermionic antisymmetry.

HF replaces this product by a normalized determinant of orthonormal spin orbitals:

$$\boxed{\Phi_{\text{HF}}(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det[\chi_i(x_j)]_{i,j=1}^{N}, \quad \langle \chi_i \mid \chi_j \rangle = \delta_{ij}.}$$

Interchanging two electronic coordinates exchanges two columns and changes the sign. The orbitals are variational unknowns, not automatically exact eigenfunctions of the isolated one-electron Hamiltonian h.

4. Derivation of the Hartree–Fock equations

For the determinant in Question 3, the energy is

$$E_{\mathrm{HF}} = \sum_i h_{ii} + \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}),$$

where

$$\begin{aligned} h_{ii} &= \int \chi_i^*(x) h \chi_i(x)\,\mathrm{d}x, \\ J_{ij} &= \iint \frac{\kappa|\chi_i(x)|^2|\chi_j(x')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x\,\mathrm{d}x', \\ K_{ij} &= \iint \frac{\kappa\chi_i^*(x)\chi_j^*(x')\chi_j(x)\chi_i(x')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x\,\mathrm{d}x'. \end{aligned}$$

The direct term J_{ij} comes from the charge densities; K_{ij} is the exchange term arising from antisymmetry. For $i = j$, they cancel exactly.

Minimize with all orthonormality constraints:

$$\mathcal{L} = E_{\mathrm{HF}} - \sum_{ij}\lambda_{ij}\big(\langle\chi_i \mid \chi_j\rangle - \delta_{ij}\big).$$

The variation with respect to $\chi_i^*(x)$ gives

$$F\chi_i = \sum_j \lambda_{ij}\chi_j, \qquad F = h + \sum_{j\in\mathrm{occ}}(J_j - K_j),$$

where the operators act on an arbitrary spin orbital f as

$$\begin{aligned} (J_j f)(x) &= \left[\int \frac{\kappa|\chi_j(x')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'\right] f(x), \\ (K_j f)(x) &= \chi_j(x)\int \frac{\kappa\chi_j^*(x')f(x')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'. \end{aligned}$$

J_j is a local multiplication operator, whereas K_j is nonlocal. The Hermitian multiplier matrix can be diagonalized by a unitary rotation of occupied orbitals, which changes the determinant only by an overall phase. In the resulting canonical orbitals,

$$\boxed{\left[h + \sum_j (J_j - K_j)\right]\chi_i = \epsilon_i\chi_i.}$$

The equations are nonlinear because F depends on its occupied solutions. Guess orbitals, build F, solve, update occupations and density, and iterate to self-consistency. Compare distinct converged solutions if seeking the lowest HF energy. The total energy is not simply the orbital-energy sum:

$$E_{\mathrm{HF}} = \sum_i \epsilon_i - \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}).$$

5. The Hohenberg–Kohn functional

For a physically admissible density n with $\int n\,\mathrm{d}^3r = N$, a precise constrained-search form of the universal functional is

$$\boxed{F[n] = \min_{\Psi\to n}\langle\Psi|T + V_{ee}|\Psi\rangle,}$$

where the search is over normalized antisymmetric N-electron states with density n. The ensemble version can be used where required. "Universal" means independent of the external scalar potential for the same particle interaction and kinetic operator.

For a specified external potential,

$$\boxed{E_v[n] = F[n] + \int v_{\text{ext}}(\boldsymbol{r}) n(\boldsymbol{r})\, \mathrm{d}^3 r, \qquad E_0 = \min_{n,\, \int n = N} E_v[n].}$$

At the exact ground-state density n_0, $E_v[n_0] = E_0$. The additive fixed-nuclei constant E_{nn} may be included in total energies without changing the minimizer. The name "HK functional" is sometimes applied to $F[n]$ and sometimes to the full $E_v[n]$; specifying both avoids ambiguity. HK theory justifies a density-based variational principle; it does not provide a closed exact formula for $F[n]$.

6. Kohn–Sham equations from the density functional

Introduce the kinetic functional of a noninteracting auxiliary system, $T_s[n]$, and decompose

$$F[n] = T_s[n] + E_H[n] + E_{xc}[n], \qquad E_H[n] = \frac{\kappa}{2} \iint \frac{n(\boldsymbol{r}) n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\, \mathrm{d}^3 r\, \mathrm{d}^3 r'.$$

Thus $E_{xc} = F - T_s - E_H$ includes both nonclassical interaction effects and the difference between interacting and noninteracting kinetic energies. With occupied normalized spin orbitals φ_i and unit occupations for simplicity,

$$n(\boldsymbol{r}) = \sum_{i \in \text{occ}} \sum_{\sigma} |\varphi_i(\boldsymbol{r}, \sigma)|^2, \qquad T_s = \sum_i \langle \varphi_i | -\frac{\hbar^2 \nabla^2}{2 m_e} | \varphi_i \rangle.$$

Vary the energy

$$T_s + E_H + E_{xc} + \int v_{\text{ext}} n\, \mathrm{d}^3 r - \sum_{ij} \lambda_{ij} (\langle \varphi_i \mid \varphi_j \rangle - \delta_{ij})$$

with respect to φ_i^*. Since

$$v_H(\boldsymbol{r}) = \frac{\delta E_H}{\delta n(\boldsymbol{r})} = \int \frac{\kappa n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\, \mathrm{d}^3 r', \qquad v_{xc}(\boldsymbol{r}) = \frac{\delta E_{xc}}{\delta n(\boldsymbol{r})},$$

and $\delta n / \delta \varphi_i^* = \varphi_i$, diagonalizing the multiplier matrix yields

$$\boxed{\left[-\frac{\hbar^2 \nabla^2}{2 m_e} + v_{\text{ext}}(\boldsymbol{r}) + v_H(\boldsymbol{r}) + v_{xc}(\boldsymbol{r}) \right] \varphi_i = \epsilon_i \varphi_i.}$$

Here the terms are auxiliary kinetic energy, external attraction, classical Hartree repulsion, and exchange–correlation potential. Solve self-consistently because the last two depend on the occupied-orbital density. For a spin-polarized calculation the corresponding spin-density functional gives separate $v_{xc,\sigma}$.

With an exact functional, KS theory can reproduce the exact ground-state density and total energy; the auxiliary determinant is not generally the exact interacting many-body wavefunction, and the orbital eigenvalues are not generally all physical excitation energies. In practice the approximation to E_{xc} is a separate source of error.

7. Meaning of variational collapse in a valence calculation

Suppose a calculation is intended to find a valence eigenstate of an all-electron one-particle Hamiltonian, while lower core states are already occupied and treated separately. Minimizing the Rayleigh quotient over

an unconstrained plane-wave expansion permits admixture of these lower states. For normalized orthogonal core and valence eigenstates,

$$|\eta_\theta\rangle = \cos\theta|v\rangle + \sin\theta|c\rangle, \quad E_c < E_v,$$

$$\langle\eta_\theta|H|\eta_\theta\rangle = E_v + (E_c - E_v)\sin^2\theta.$$

The energy decreases by increasing the core component. The variational procedure therefore leaves the intended valence subspace and collapses toward an occupied core solution unless core orthogonality is imposed. Highly localized core structure also requires many high-momentum plane waves, making a naive expansion inefficient.

This does **not** mean that the ordinary nonrelativistic Coulomb Hamiltonian is unbounded below or that a correct all-electron Rayleigh–Ritz calculation is invalid. If all lower eigenstates are included and the required higher eigenstate is properly targeted, plane waves converge in principle. The issue here is the missing valence-subspace constraint, not an energy tending to $-\infty$.

8. Orthogonalized plane waves

Let $\{|\chi_{c\boldsymbol{k}}\rangle\}$ be orthonormal core Bloch states at the chosen $\boldsymbol{k}$. Define

$$P_c = \sum_c |\chi_{c\boldsymbol{k}}\rangle\langle\chi_{c\boldsymbol{k}}|, \qquad Q = 1 - P_c.$$

Replace each plane wave by its component orthogonal to the occupied core:

$$\boxed{|\widetilde{w}_{\boldsymbol{k}+\boldsymbol{G}}\rangle = Q|\boldsymbol{k}+\boldsymbol{G}\rangle = |\boldsymbol{k}+\boldsymbol{G}\rangle - \sum_c |\chi_{c\boldsymbol{k}}\rangle\langle\chi_{c\boldsymbol{k}} \mid \boldsymbol{k}+\boldsymbol{G}\rangle.}$$

Indeed $\langle\chi_{d\boldsymbol{k}}|Q = 0$, so every linear combination remains orthogonal to all core states. These are orthogonalized plane waves (OPWs). They need not be mutually orthonormal: their overlap matrix is

$$S_{\boldsymbol{G}\boldsymbol{G}'} = \delta_{\boldsymbol{G}\boldsymbol{G}'} - \sum_c \langle\boldsymbol{k}+\boldsymbol{G} \mid \chi_{c\boldsymbol{k}}\rangle\langle\chi_{c\boldsymbol{k}} \mid \boldsymbol{k}+\boldsymbol{G}'\rangle.$$

Null or linearly dependent combinations must be removed before solving the generalized eigenproblem.

9. Physical and pseudo wavefunctions

Expand a smooth auxiliary function in ordinary plane waves,

$$\varphi_{\boldsymbol{k}}(\boldsymbol{r}) = \sum_{\boldsymbol{G}} a_{\boldsymbol{G}} \frac{\mathrm{e}^{\mathrm{i}(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}}{\sqrt{\Omega}}.$$

The physical OPW state is

$$\boxed{|\psi_{\boldsymbol{k}}\rangle = Q|\varphi_{\boldsymbol{k}}\rangle = |\varphi_{\boldsymbol{k}}\rangle - \sum_c |\chi_{c\boldsymbol{k}}\rangle\langle\chi_{c\boldsymbol{k}} \mid \varphi_{\boldsymbol{k}}\rangle.}$$

Equivalently $\varphi = \psi + P_c\varphi$. The pseudo function is not uniquely determined by ψ: any added core component is removed by Q. A useful representative is chosen to be smooth. Since ψ is already orthogonal to the core, writing $\varphi = Q\psi$ would merely give $\varphi = \psi$ and would not express the intended construction. Normalization is $\langle\psi \mid \psi\rangle = \langle\varphi|Q|\varphi\rangle = 1$; φ need not have unit ordinary norm.

10. Effective Hamiltonian and plane-wave solution

(a) Phillips–Kleinman form

Assume core eigenstates satisfy $H|\chi_c\rangle = \epsilon_c|\chi_c\rangle$, with a Hermitian H. From $H\psi = \epsilon\psi$ and $\psi = (1 - P_c)\varphi$,

$$H\varphi - \sum_c \epsilon_c|\chi_c\rangle\langle\chi_c \mid \varphi\rangle = \epsilon\varphi - \sum_c \epsilon|\chi_c\rangle\langle\chi_c \mid \varphi\rangle.$$

Rearranging gives

$$\boxed{H^{(\mathrm{eff})}(\epsilon) = H + V_R(\epsilon), \qquad V_R(\epsilon) = \sum_c (\epsilon - \epsilon_c)|\chi_c\rangle\langle\chi_c|,}$$

$$H^{(\mathrm{eff})}(\epsilon)\varphi = \epsilon\varphi.$$

For a valence energy above all relevant core levels,

$$\langle f|V_R|f\rangle = \sum_c (\epsilon - \epsilon_c)|\langle\chi_c \mid f\rangle|^2 \geq 0.$$

The extra nonlocal, energy-dependent repulsion accounts for core orthogonality and counteracts the deep attractive potential in the core region. Writing $H = T + V$ identifies $V_{\mathrm{ps}} = V + V_R$. This exact projector construction should not be confused with an arbitrary local potential independent of energy.

(b) Matrix equations and convergence

Keep reciprocal vectors satisfying a plane-wave kinetic cutoff, expand φ as in Question 9, and project onto each retained plane wave. For a local periodic V, the matrix equation is

$$\sum_{\boldsymbol{G}'} \left[\frac{\hbar^2|\boldsymbol{k} + \boldsymbol{G}|^2}{2m_e}\delta_{\boldsymbol{G}\boldsymbol{G}'} + V_{\boldsymbol{G}-\boldsymbol{G}'} + (V_R)_{\boldsymbol{G}\boldsymbol{G}'}(\epsilon) \right] a_{\boldsymbol{G}'} = \epsilon a_{\boldsymbol{G}}.$$

Here $V_{\boldsymbol{G}-\boldsymbol{G}'}$ is the Fourier coefficient of the local potential, and

$$(V_R)_{\boldsymbol{G}\boldsymbol{G}'}(\epsilon) = \sum_c (\epsilon - \epsilon_c)\langle\boldsymbol{k} + \boldsymbol{G} \mid \chi_c\rangle\langle\chi_c \mid \boldsymbol{k} + \boldsymbol{G}'\rangle.$$

Because the exact projector potential depends on the unknown energy, this is not simply a fixed ordinary matrix eigenproblem. Moving its ϵ terms to the right gives the equivalent OPW generalized eigenproblem

$$\boxed{\sum_{\boldsymbol{G}'} H^{\mathrm{OPW}}_{\boldsymbol{G}\boldsymbol{G}'} a_{\boldsymbol{G}'} = \epsilon \sum_{\boldsymbol{G}'} S_{\boldsymbol{G}\boldsymbol{G}'} a_{\boldsymbol{G}'},}$$

$$H^{\mathrm{OPW}}_{\boldsymbol{G}\boldsymbol{G}'} = \langle\boldsymbol{k} + \boldsymbol{G}|QHQ|\boldsymbol{k} + \boldsymbol{G}'\rangle, \quad S_{\boldsymbol{G}\boldsymbol{G}'} = \langle\boldsymbol{k} + \boldsymbol{G}|Q|\boldsymbol{k} + \boldsymbol{G}'\rangle.$$

Solve in the nonsingular retained subspace, increase the cutoff to test convergence, and sample the required $\boldsymbol{k}$ points. Modern transferable pseudopotentials replace the exact energy-dependent construction by a separately generated representation, often with nonlocal angular-momentum channels. A self-consistent electronic potential requires an additional density-update loop.

11. Why sodium's valence binding is not hydrogen's $1s$ energy

Far outside a neutral sodium atom's ten-electron core, the remaining ion has net charge $+e$, so the valence electron sees a $-\kappa/r$ tail. This does *not* make the potential equal to $-\kappa/r$ at all radii. Core screening, exchange, penetration, and orthogonality to occupied core orbitals change the short-range problem.

There are two separate errors in identifying Na $3s$ with H $1s$:

(i) Even in a purely hydrogenic charge-$+1$ potential, the $n = 3$ state would have $E_3 = -13.6/9 \simeq -1.51\,\text{eV}$, not $-13.6\,\text{eV}$. The latter is the $n = 1$ ground state of that different Hamiltonian.

(ii) Removing the core nodes to construct a nodeless pseudo s orbital does not turn it into the hydrogen $1s$ problem. Its Hamiltonian contains the core-induced effective repulsion derived in Question 10, which modifies the short-range binding while preserving the valence energy.

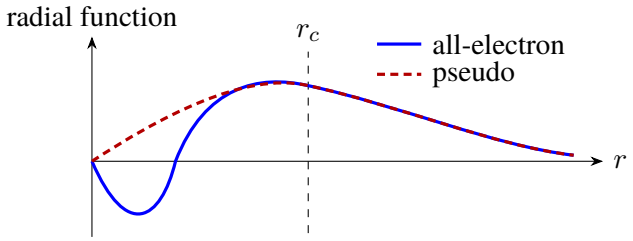
The observed first ionization energy of Na is approximately $5.1391\,\text{eV}$, so the removal energy relative to the continuum is approximately $-5.14\,\text{eV}$ in the convention used by the question. The long-range tail alone cannot calculate this value; a realistic core potential or an equivalent fitted scattering/quantum-defect description is needed. In a many-body interpretation, an experimental ionization energy is a total-energy difference, not automatically an arbitrary approximate HF or KS orbital eigenvalue.

As a numerical characterization, writing $E \simeq -13.6057\,\text{eV}/(n - \delta_s)^2$ gives an effective $n^* \simeq 1.627$ and $\delta_s \simeq 1.373$ for the Na $3s$ binding. This encodes a short-range core effect; it is not derived merely by replacing the nuclear charge by one.

12. Pseudo atom

A pseudo atom is an auxiliary atomic problem in which the nucleus and tightly bound core electrons are replaced by an effective ionic pseudopotential, while valence electrons are treated explicitly. For Na, this commonly means an effective ten-electron core and one explicit valence electron; a particular pseudopotential may choose a different valence partition.

Within a selected core radius, pseudo radial orbitals are smooth and usually nodeless in the valence channel. Outside that radius they are constructed to reproduce the chosen all-electron valence behavior, with reference energies and scattering properties matched; norm-conserving constructions also match the valence norm inside the core. A pseudo atom is a computational replacement with a specified transferability range, not a new chemical element.



Schematic only: smooth replacement inside r_c, matching outside. The number of depicted inner nodes is illustrative, not a computed Na $3s$ radial solution.

Primary references used for checks

- P. Hohenberg and W. Kohn, Phys. Rev. 136, B864 (1964) and W. Kohn and L. J. Sham, Phys. Rev. 140, A1133 (1965): the density-functional and auxiliary-orbital framework.

- J. C. Phillips and L. Kleinman, "New Method for Calculating Wave Functions in Crystals and Molecules," Phys. Rev. 116, 287 (1959): OPW orthogonality and the effective repulsive potential.
- NIST elemental data for sodium. The ground configuration is $[\mathrm{Ne}]3s$. The first ionization energy is $5.13907696\,\mathrm{eV}$.

Chapter 6

Solid State Theory: April 26, 2017 — Solutions

These solutions are independently derived, not an official answer key. Reciprocal vectors satisfy $\mathbf{g}_i \cdot \mathbf{t}_j = 2\pi\delta_{ij}$. Unless otherwise stated, a density of states is per unit area, per spin, and for the specified band or patch.

1. Translation symmetry and the first Brillouin zone

(a) False. Atomic positions in a crystal have the general form

$$\mathbf{r}_{\ell\alpha} = \mathbf{R}_\ell + \mathbf{d}_\alpha, \qquad \mathbf{R}_\ell = \sum_i n_i \mathbf{t}_i.$$

Atoms occupying the same basis site α in different cells are related by a Bravais translation. However, atoms of the same chemical species can occupy different basis sites. Diamond provides a counterexample: an FCC Bravais lattice with basis $\mathbf{d}_1 = 0$ and $\mathbf{d}_2 = (a/4)(1,1,1)$. Both sites are carbon, but their separation is not an FCC translation. Chemical identity does not imply translational equivalence.

(b) True. The first Brillouin zone is the reciprocal-space Wigner-Seitz cell about the origin:

$$\mathcal{B}_1 = \{\mathbf{k} : \ |\mathbf{k}| \le |\mathbf{k} - \mathbf{G}| \text{ for every reciprocal vector } \mathbf{G}\}.$$

A point outside this closed region violates at least one inequality. Thus some $\mathbf{G} \neq 0$ satisfies $|\mathbf{k} - \mathbf{G}| < |\mathbf{k}|$. Boundary points can be equidistant from two reciprocal sites; they are not strictly outside the closed first zone.

2. FCC plane spacing and planar density

The primitive volume and reciprocal vectors are

$$\Omega = \mathbf{t}_1 \cdot (\mathbf{t}_2 \times \mathbf{t}_3) = \frac{a^3}{4},$$

$$\mathbf{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \mathbf{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \mathbf{g}_3 = \frac{2\pi}{a}(1,1,-1).$$

Their dot products verify $\mathbf{g}_i \cdot \mathbf{t}_j = 2\pi\delta_{ij}$. If $\mathbf{G} = \sum_i m_i \mathbf{g}_i$ with relatively prime integer coefficients, then

$$\mathbf{G} \cdot \mathbf{R} = 2\pi \sum_i m_i n_i = 2\pi\ell, \qquad \ell \in \mathbb{Z}.$$

Because $\gcd(m_1, m_2, m_3) = 1$, every integer ℓ occurs. Successive planes containing lattice points therefore have spacing $2\pi/|\mathbf{G}|$. A merely parallel vector cannot be substituted without checking its length and integer coefficients.

(a) A normal with half-integer reciprocal coefficients

The shortest nonzero reciprocal vector parallel to $\mathbf{g}_1/2 + \mathbf{g}_2 - \mathbf{g}_3$ is

$$\mathbf{G} = \mathbf{g}_1 + 2\mathbf{g}_2 - 2\mathbf{g}_3 = \frac{2\pi}{a}(-1, -3, 5).$$

The coefficients $(1, 2, -2)$ are relatively prime, giving

$$\boxed{d = \frac{2\pi}{|\mathbf{G}|} = \frac{a}{\sqrt{35}}.}$$

(b) Number of lattice points per unit plane area

On the plane through the origin, $n_1 + 2n_2 - 2n_3 = 0$. Every in-plane lattice vector is uniquely

$$\mathbf{R} = n_2 \underbrace{(-2\mathbf{t}_1 + \mathbf{t}_2)}_{\mathbf{u}} + n_3 \underbrace{(2\mathbf{t}_1 + \mathbf{t}_3)}_{\mathbf{v}}.$$

This proves that $\mathbf{u}, \mathbf{v}$ form a primitive planar basis, not a supercell. In Cartesian coordinates,

$$\mathbf{u} = \frac{a}{2}(1, -2, -1), \quad \mathbf{v} = \frac{a}{2}(1, 3, 2), \quad \mathbf{u} \times \mathbf{v} = \frac{a^2}{4}(-1, -3, 5).$$

The planar cell has area $A_\parallel = a^2\sqrt{35}/4$ and contains one lattice point. Therefore

$$\boxed{\sigma = \frac{1}{A_\parallel} = \frac{4}{a^2\sqrt{35}}.}$$

As a check, $A_\parallel d = \Omega$, so $\sigma = d/\Omega$. This counts Bravais lattice points. Multiplying again by the four points in an FCC conventional cell would count the cell conversion twice.

(c) A normal in the direct-lattice basis

First convert the specified normal to Cartesian coordinates:

$$\mathbf{N} = \mathbf{t}_1 + 2\mathbf{t}_2 - 2\mathbf{t}_3 = \frac{a}{2}(0, -1, 3).$$

For a reciprocal vector $\mathbf{G}' \parallel \mathbf{N}$, the integer coefficients have ratio

$$m_1 : m_2 : m_3 = (\mathbf{N} \cdot \mathbf{t}_1) : (\mathbf{N} \cdot \mathbf{t}_2) : (\mathbf{N} \cdot \mathbf{t}_3) = 2 : 3 : -1.$$

The shortest such reciprocal vector is

$$\mathbf{G}' = 2\mathbf{g}_1 + 3\mathbf{g}_2 - \mathbf{g}_3 = \frac{2\pi}{a}(0, -2, 6), \qquad \boxed{d' = \frac{a}{2\sqrt{10}}.}$$

Independently, the projection of $\mathbf{R} = (a/2)(n_2 + n_3, n_1 + n_3, n_1 + n_2)$ onto $(0, -1, 3)/\sqrt{10}$ is $a(2n_1 + 3n_2 - n_3)/(2\sqrt{10})$. The numerator takes every integer value. The vector $(2\pi/a)(0, -1, 3)$ is not an FCC reciprocal lattice vector; using it would miss the factor of two.

3. Critical points and local density of states

The stated two-dimensional graphite model describes the two π bands of a honeycomb layer. Set

$$x = ak_x, \quad y = ak_y, \quad h(x,y) = |F_{\mathbf{k}}|^2 = 1 + 4\cos(x/2)\cos(\sqrt{3}y/2) + 4\cos^2(x/2).$$

Where $h > 0$, $\nabla\epsilon_\pm = \pm\gamma\nabla h/(2\sqrt{h})$. At $h = 0$, this formula cannot be evaluated by interpreting $0/0$ as zero.

(a) Expansions at the three points

Γ: smooth extrema. With $\mathbf{q} = \mathbf{k}$,

$$h = 9 - \frac{3}{2}a^2(q_x^2 + q_y^2) + O(a^4q^4), \qquad |F| = 3 - \frac{1}{4}a^2q^2 + O(a^4q^4).$$

Thus, with $\alpha = \gamma a^2/4 > 0$,

$$\epsilon_+ = \epsilon_p + 3\gamma - \alpha q^2 + O(q^4), \qquad \epsilon_- = \epsilon_p - 3\gamma + \alpha q^2 + O(q^4).$$

Both gradients vanish. The upper band has a local maximum, a negative-definite Hessian, and Morse index 2; the lower band has a local minimum, a positive-definite Hessian, and index 0.

Q: a saddle in each band. At $(x,y) = (\pi, \pi/\sqrt{3})$,

$$h = 1, \quad \nabla h = 0, \quad H_h = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix}.$$

Its eigenvalues are $3, -1$, with orthonormal directions $\hat{\mathbf{u}} = (\sqrt{3}/2, 1/2)$ and $\hat{\mathbf{v}} = (-1/2, \sqrt{3}/2)$. For displacement components q_u, q_v,

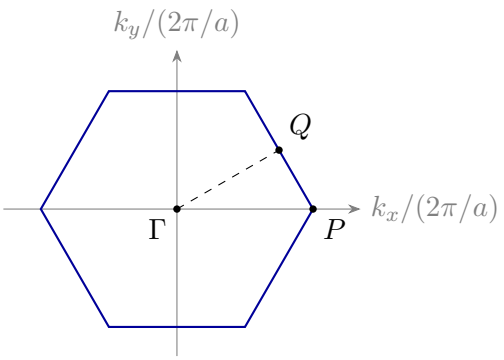
$$\epsilon_\pm = \epsilon_p \pm \gamma \pm \frac{\gamma a^2}{4}(3q_u^2 - q_v^2) + O(a^3q^3).$$

The curvatures have opposite signs, so both bands have saddles of index 1. Being on a zone boundary does not invalidate this local classification: the dispersion extends periodically.

P: a Dirac contact, not an ordinary smooth stationary point. At $(x,y) = (4\pi/3, 0)$, $h = 0$, while

$$h = \frac{3}{4}a^2q^2 + O(a^3q^3), \qquad \epsilon_\pm = \epsilon_p \pm c|\mathbf{q}| + O(q^2), \qquad c = \frac{\sqrt{3}\gamma a}{2}.$$

The upper band has a conical local minimum and the lower band a conical local maximum, and the bands touch. Each energy-ordered band is nondifferentiable at the cone tip. Its gradient does not exist; it is not a smooth van Hove or Morse stationary point satisfying $\nabla\epsilon = 0$. If the term critical point includes band-degeneracy singularities, identify P explicitly as a *Dirac band-contact singularity*. The directional slopes are finite, not infinite. A quadratic-extremum DOS formula is inapplicable.



The reconstructed first Brillouin zone places Q at an edge midpoint and P at a corner, using the question's coordinates.

(b) Choosing Γ: direct evaluation of the local DOS

Choose a disk $q < \Lambda \ll a^{-1}$ and let $E_{\min} = \epsilon_p - 3\gamma$. The number of states per unit sample area per k-space area is $(2\pi)^{-2}$. In the quadratic approximation,

$$\begin{aligned} g_{\Gamma,-}(E) &= \frac{1}{(2\pi)^2}\int_{q<\Lambda} \mathrm{d}^2 q\, \delta(E - E_{\min} - \alpha q^2) \\ &= \frac{1}{2\pi}\int_0^{\Lambda} q\ \mathrm{d}q\, \delta(E - E_{\min} - \alpha q^2) \\ &= \boxed{\frac{1}{\pi\gamma a^2}\Theta(E - E_{\min})\Theta(E_{\min} + \alpha\Lambda^2 - E).} \end{aligned}$$

This is the step onset of a two-dimensional quadratic minimum. For the upper band,

$$g_{\Gamma,+}(E) = \frac{1}{\pi\gamma a^2}\Theta(E_{\max} - E)\Theta(E - E_{\max} + \alpha\Lambda^2), \quad E_{\max} = \epsilon_p + 3\gamma.$$

These are approximate contributions from the chosen patch, not the complete band DOS. Multiply by $A_c = \sqrt{3}a^2/2$ for DOS per primitive cell, by 2 for both spins, or by sample area A for total DOS. Integrating either expression over energy gives $\Lambda^2/(4\pi)$, the number of states in the chosen disk per unit sample area.

For comparison, one Q saddle produces a logarithmic singularity. One P valley, with its upper and lower conical branches, gives

$$g_P(E) = \frac{|E - \epsilon_p|}{2\pi c^2} = \frac{2|E - \epsilon_p|}{3\pi\gamma^2 a^2}.$$

Spin and the second inequivalent valley have not been included. The DOS vanishes at the Dirac point; it is not the constant DOS of a two-dimensional parabolic band edge.

4. Bloch's theorem

For a one-electron Hamiltonian with $V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$, energy eigenstates can be chosen as

$$\boxed{\psi_{n\mathbf{k}}(\mathbf{r}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \qquad u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r}).}$$

Equivalently, $\psi_{n\mathbf{k}}(\mathbf{r}+\mathbf{R}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}}\psi_{n\mathbf{k}}(\mathbf{r})$. The Hamiltonian commutes with all lattice translations, which also commute with one another. Simultaneous eigenstates may be chosen; the translation-group multiplication law fixes their translation eigenvalues to the displayed phase form. Here n is a band index, and $\mathbf{k}, \mathbf{k} + \mathbf{G}$ are equivalent quasimomentum labels. The Bloch wave itself need not be periodic; its probability density and its periodic factor $u_{n\mathbf{k}}$ are periodic.

5. Why exchange lowers the interaction energy

Let $x = (\mathbf{r}, \sigma)$ include spin and let $w(\mathbf{r} - \mathbf{r}') = e^2/(4\pi\varepsilon_0|\mathbf{r} - \mathbf{r}'|)$. For a fixed orthonormal set of spin-orbitals,

$$\begin{aligned} J_{ij} &= \int \mathrm{d}x\ \mathrm{d}x'\, |\chi_i(x)|^2 w(\mathbf{r} - \mathbf{r}')|\chi_j(x')|^2, \\ K_{ij} &= \int \mathrm{d}x\ \mathrm{d}x'\, \chi_i^*(x)\chi_j(x) w(\mathbf{r} - \mathbf{r}')\chi_j^*(x')\chi_i(x'). \end{aligned}$$

A Hartree product lacks exchange cross terms. Antisymmetrizing into a Slater determinant gives

$$E_{ee}^{\mathrm{HF}} = \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}).$$

Since $J_{ii} = K_{ii}$, self-interaction cancels. Relative to the same-orbital Hartree-product interaction energy $E_{ee}^{\mathrm{H}} = \frac{1}{2}\sum_{i\neq j} J_{ij}$,

$$E_{ee}^{\mathrm{HF}} - E_{ee}^{\mathrm{H}} = -\frac{1}{2}\sum_{i\neq j} K_{ij} \leq 0.$$

The Coulomb kernel is positive definite, so $K_{ij} \geq 0$; exchange between opposite-spin functions vanishes. If the reference is instead the full-density classical Hartree energy $\frac{1}{2}\sum_{ij} J_{ij}$, the usual exchange functional is $E_x = -\frac{1}{2}\sum_{ij} K_{ij}$, including the diagonal self-interaction correction.

Antisymmetry reduces the likelihood of same-spin electrons being close. This exchange hole reduces repulsion; no extra attractive force is introduced. The comparison fixes the orbitals and the Hartree reference. It does not prove universal strict ordering of separately optimized energies in every Hartree and HF variational convention.

6. The Hohenberg-Kohn functional

Write $\hat{H}_v = \hat{T} + \hat{W}_{ee} + \sum_i v(\mathbf{r}_i)$. Under the usual nondegenerate-ground-state assumptions, the ground-state density determines the external potential up to a constant, and hence ground-state properties. A constrained-search expression for the universal functional is

$$\boxed{F[n] = \min_{\Psi\to n} \langle\Psi|\hat{T} + \hat{W}_{ee}|\Psi\rangle.}$$

The search is over normalized antisymmetric N-electron wavefunctions with density n. It contains kinetic and electron-electron energies but no explicit external potential. The latter enters separately:

$$E_v[n] = F[n] + \int v(\mathbf{r})n(\mathbf{r})\ \mathrm{d}^3r, \qquad E_0 = \min_{\substack{n \text{ admissible} \\ \int n = N}} E_v[n].$$

The minimum occurs at the physical ground-state density. Where the functional derivative exists, the Euler equation is $\delta F/\delta n(\mathbf{r}) + v(\mathbf{r}) = \mu$, with μ enforcing electron number. Universality does not imply that different potentials have the same minimizing density: changing $\int vn$ changes the minimum. Existence of the exact functional does not provide a practical explicit formula for it.

7. The local density approximation

The Kohn-Sham decomposition is

$$F[n] = T_s[n] + E_H[n] + E_{xc}[n], \qquad E_H[n] = \frac{1}{2}\int \mathrm{d}^3r\, \mathrm{d}^3r'\, n(\mathbf{r})w(\mathbf{r}-\mathbf{r}')n(\mathbf{r}').$$

T_s is the kinetic energy of the auxiliary noninteracting system with the same density. E_{xc} includes the interaction correction beyond classical Hartree energy and the difference between interacting kinetic energy and T_s. LDA uses the uniform-gas exchange-correlation energy *per electron* at the local density:

$$\boxed{E_{xc}^{\mathrm{LDA}}[n] = \int n(\mathbf{r})\varepsilon_{xc}^{\mathrm{unif}}(n(\mathbf{r}))\ \mathrm{d}^3r.}$$

The factor n is required because $\varepsilon_{xc}^{\mathrm{unif}}$ is per electron. Its derivative gives

$$v_{xc}^{\mathrm{LDA}}(\mathbf{r}) = \left[\varepsilon_{xc}^{\mathrm{unif}}(n) + n\frac{\mathrm{d}\varepsilon_{xc}^{\mathrm{unif}}}{\mathrm{d}n}\right]_{n=n(\mathbf{r})}.$$

LDA treats an inhomogeneous system as locally uniform. It is motivated for slowly varying densities, omits explicit gradients and general nonlocal information, and is not exact for arbitrary systems. Dependence on $n_\uparrow, n_\downarrow$ gives the local spin-density approximation.

8. Hartree, Hartree-Fock, and Kohn-Sham equations

The common one-body operator is

$$\hat{h}_0 = -\frac{\hbar^2}{2m}\nabla^2 + v_{\text{ext}}(\mathbf{r}).$$

The terms are electron kinetic energy and the external potential from fixed nuclei and other external sources. Periodic calculations use Bloch orbitals and sums over occupied **k** points.

Hartree: the product-state mean field

$$\boxed{\left[\hat{h}_0 + \sum_{j\neq i} J_j(\mathbf{r})\right]\chi_i(x) = \varepsilon_i\chi_i(x), \quad J_j(\mathbf{r}) = \int \mathrm{d}x'\, w(\mathbf{r}-\mathbf{r}')|\chi_j(x')|^2.}$$

Each electron experiences the other electrons' average Coulomb potential. The restriction $j \neq i$ excludes its own orbital's self-interaction. The approximation omits exchange from antisymmetry and genuine many-electron correlation.

Hartree-Fock: a single Slater determinant

$$\boxed{\left[\hat{h}_0 + \sum_j (\hat{J}_j - \hat{K}_j)\right]\chi_i = \varepsilon_i\chi_i,}$$

where $\hat{J}_j\chi_i = J_j(\mathbf{r})\chi_i(x)$ and

$$(\hat{K}_j\chi_i)(x) = \chi_j(x)\int \mathrm{d}x'\, w(\mathbf{r}-\mathbf{r}')\chi_j^*(x')\chi_i(x').$$

The exchange operator is nonlocal: its value at x depends on orbital values elsewhere. For $j = i$, exchange cancels the direct term. HF optimizes single determinants, not the full interacting many-electron Hilbert space.

Kohn-Sham: an auxiliary system reproducing the density

$$\boxed{\left[-\frac{\hbar^2}{2m}\nabla^2 + v_{\text{ext}}(\mathbf{r}) + v_H(\mathbf{r}) + v_{xc}(\mathbf{r})\right]\phi_i = \epsilon_i\phi_i,}$$

$$v_H(\mathbf{r}) = \int \mathrm{d}^3r'\, w(\mathbf{r}-\mathbf{r}')n(\mathbf{r}'), \quad v_{xc}(\mathbf{r}) = \frac{\delta E_{xc}}{\delta n(\mathbf{r})}, \quad n(\mathbf{r}) = \sum_i f_i|\phi_i(\mathbf{r})|^2.$$

Use spin-orbitals with $f_i \leq 1$, or unpolarized spatial orbitals with $f_i \leq 2$, without counting spin twice. The orbital kinetic energy T_s generally differs from the interacting kinetic energy; the difference is included in E_{xc}. The electronic total energy is

$$\begin{aligned} E &= T_s + \int v_{\text{ext}} n\ \mathrm{d}^3r + E_H + E_{xc} \\ &= \sum_i f_i\epsilon_i - E_H + E_{xc} - \int n v_{xc}\ \mathrm{d}^3r. \end{aligned}$$

Summing orbital eigenvalues alone double-counts mean-field contributions. Add fixed nuclear repulsion E_{NN} if the total electron-plus-nuclei energy is required.

Self-consistent solution

All three problems are nonlinear because their effective operators depend on the unknown orbitals or density. Choose a basis or grid and boundary conditions; guess orbitals or a density; build the direct, exchange, or exchange-correlation terms; solve the eigenvalue problem; occupy states for the required electron number; compute and usually mix the new density; rebuild and repeat. Check density, total-energy, and equation-residual convergence, followed by basis, grid, and $\mathbf{k}$-mesh convergence. Different initial densities or spin states can reach different stationary solutions. SCF convergence alone does not establish a global minimum.

Primary references and verification scope

The graphite model is discussed in P. R. Wallace, The Band Theory of Graphite (1947). The expansions, lattice-plane calculations, and DOS normalization above follow directly from the question data. Density-functional foundations are given in Hohenberg and Kohn, Inhomogeneous Electron Gas (1964) and Kohn and Sham, Self-Consistent Equations Including Exchange and Correlation Effects (1965).

Chapter 7

Solutions: Midterm Exam, November 8, 2017

The density of states $D(E)$ below counts states per unit energy in the entire sample of volume V. A factor g_s explicitly specifies spin counting. We use $g_s = 2$ for the unpolarized free-electron gas.

1. A change of primitive basis

Let $T = (\boldsymbol{t}_1\ \boldsymbol{t}_2\ \boldsymbol{t}_3)$ and $S = (\boldsymbol{s}_1\ \boldsymbol{s}_2\ \boldsymbol{s}_3)$. The stated relations give $S = TA$, where

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 1 & \alpha \\ 2 & \alpha & 1 \end{pmatrix}, \qquad \det A = -(2\alpha+1)(\alpha+1) = -2\alpha^2 - 3\alpha - 1.$$

Two primitive bases generate the same lattice precisely when their change-of-basis matrix is *integer unimodular*: A has integer entries and $\det A = \pm 1$. Necessity follows because both sets of basis vectors must be integer combinations of the other set. Sufficiency follows from the adjugate formula: an integer matrix with determinant ± 1 has an integer inverse.

The determinant equation $\det A = +1$ gives $2\alpha^2 + 3\alpha + 2 = 0$, whose discriminant is -7. The equation $\det A = -1$ gives

$$\alpha(2\alpha + 3) = 0, \qquad \alpha = 0 \text{ or } -\frac{3}{2}.$$

The second root gives noninteger coordinates for $\boldsymbol{s}_2$ and $\boldsymbol{s}_3$ in the original primitive basis; equal cell volumes alone do not make it a basis of the same lattice. Therefore

$$\boxed{\alpha = 0}.$$

As a direct sufficiency check, at $\alpha = 0$ the inverse relations are

$$\boldsymbol{t}_1 = -\boldsymbol{s}_1 - \boldsymbol{s}_2 + 2\boldsymbol{s}_3, \qquad \boldsymbol{t}_2 = -\boldsymbol{s}_1 + 2\boldsymbol{s}_3, \qquad \boldsymbol{t}_3 = \boldsymbol{s}_1 + \boldsymbol{s}_2 - \boldsymbol{s}_3.$$

They express every original lattice point as an integer combination of the new primitive vectors.

2. Silicon as a lattice and a basis

Crystalline silicon has the diamond structure: an fcc Bravais lattice with two identical Si atoms in each primitive cell. Using the conventional cubic lattice constant a, one may choose

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \quad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \quad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0), \qquad \boldsymbol{\tau}_1 = (0,0,0), \quad \boldsymbol{\tau}_2 = \frac{a}{4}(1,1,1).$$

The full set of atomic positions is $\boldsymbol{R} + \boldsymbol{\tau}_\nu$, with $\boldsymbol{R}$ an integer combination of the three translations. The conventional cubic cell contains four fcc lattice points and hence eight Si atoms; the primitive cell contains two.

3. The stated bcc lattice with a two-atom basis

> The source explicitly specifies *two atoms per primitive cell*. This is followed below. It is different from a monatomic bcc crystal, which has one atom per primitive cell and two per conventional cubic cell. The unspecified two-atom basis changes the atomic count, but not the Bravais-cell or Brillouin-zone volumes under the stated bcc assumption.

(a) Atomic number density

A conventional bcc cube contains $8(1/8) + 1 = 2$ lattice points. Its primitive volume is thus $\Omega = a^3/2$. With two atoms per primitive cell,

$$\boxed{n_{\text{atom}} = \frac{2}{\Omega} = \frac{4}{a^3}}.$$

If the intended premise had instead been monatomic bcc, the density would be $2/a^3$; that is not the literal premise of this question.

(b) Wigner–Seitz-cell volume

A Wigner–Seitz cell contains one Bravais lattice point and has the primitive volume:

$$\boxed{V_{\text{WS}} = \frac{a^3}{2}}.$$

(c) Brillouin-zone volume

The Brillouin zone is a reciprocal Wigner–Seitz cell, so

$$\boxed{V_{\text{BZ}} = \frac{(2\pi)^3}{\Omega} = \frac{16\pi^3}{a^3}}.$$

(d) Number of facets

The reciprocal lattice of bcc is fcc. In units $2\pi/a$, its shortest nonzero reciprocal vectors are

$$(\pm 1, \pm 1, 0), \quad (\pm 1, 0, \pm 1), \quad (0, \pm 1, \pm 1).$$

There are twelve. Each perpendicular-bisector plane $\boldsymbol{k} \cdot \boldsymbol{G} = |\boldsymbol{G}|^2/2$ supplies one face of the reciprocal Wigner–Seitz cell. The resulting polyhedron is a rhombic dodecahedron with

$$\boxed{12 \text{ facets}}.$$

The fourteen-faced truncated octahedron is the *direct-space* Wigner–Seitz cell of bcc, not its first Brillouin zone.

4. Reciprocal vectors and planar density

(a) Reciprocal primitive vectors

The direct primitive volume is $\Omega = a^3/4$. The cross-product formula gives

$$\boldsymbol{g}_1 = \frac{2\pi}{\Omega}(\boldsymbol{t}_2 \times \boldsymbol{t}_3),$$

with cyclic permutations for the other two vectors. Therefore

$$\boxed{\boldsymbol{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \boldsymbol{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \boldsymbol{g}_3 = \frac{2\pi}{a}(1,1,-1)}.$$

For instance, the first direct vector has dot product 2π with $\boldsymbol{g}_1$ and zero with $\boldsymbol{g}_2, \boldsymbol{g}_3$, verifying the required normalization.

(b) Density on successive lattice planes

Let $A_\parallel$ be the primitive area of the two-dimensional lattice on one plane, containing one lattice point. Successive occupied planes are separated by d. The primitive prism obtained from two in-plane translations and one translation to the next plane has volume $A_\parallel d = \Omega$; a lateral shift of the next layer does not affect this volume. Thus

$$\boxed{\sigma = \frac{1}{A_\parallel} = \frac{d}{\Omega} = \frac{4d}{a^3}}.$$

A second derivation counts points in a large slab: L/d layers of area A contain approximately $\sigma AL/d$ points, which must equal AL/Ω. Boundary corrections vanish in the macroscopic limit.

5. The Born–von Karman sum

Define

$$F(\boldsymbol{k}) = k_x k_y k_z \exp(-k_x^2/\alpha - k_y^2/\beta - k_z^2/\gamma).$$

The allowed wave vectors of a Born–von Karman periodic box form an inversion-symmetric set: if $\boldsymbol{k}$ is allowed, so is $-\boldsymbol{k}$. Since the Gaussian is even and the product of three components is odd,

$$F(-\boldsymbol{k}) = -F(\boldsymbol{k}).$$

Every pair cancels, and the unpaired origin contributes zero. The positive parameters ensure absolute convergence, so pairing the infinite sum is legitimate. Consequently

$$\boxed{\frac{1}{V}\sum_{\boldsymbol{k}} F(\boldsymbol{k}) = 0}.$$

This is an exact discrete cancellation, not merely a continuum approximation. In the thermodynamic limit it also follows by replacing $V^{-1}\sum_{\boldsymbol{k}}$ by $(2\pi)^{-3}\int \mathrm{d}^3k$; each Cartesian odd Gaussian integral vanishes.

6. Translating a wavefunction between cells

Put $\boldsymbol{T} = \boldsymbol{t}_m - \boldsymbol{t}_n$. A point $\boldsymbol{r}$ in the cell centered on $\boldsymbol{t}_m$ corresponds to $\boldsymbol{r} - \boldsymbol{T}$ in the cell centered on $\boldsymbol{t}_n$. Bloch's theorem says

$$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{T}}\psi_{\boldsymbol{k}}(\boldsymbol{r} - \boldsymbol{T}).$$

Since the latter function is known as f in the original cell,

$$\boxed{\psi_{\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{t}_m - \boldsymbol{t}_n)} f\big(\boldsymbol{r} - (\boldsymbol{t}_m - \boldsymbol{t}_n)\big)}.$$

Equivalently, with the same local coordinate $\boldsymbol{\rho}$ in both cells, $\psi(\boldsymbol{t}_m + \boldsymbol{\rho}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{T}} f(\boldsymbol{t}_n + \boldsymbol{\rho})$. Both the phase and the shifted function argument are required when f is expressed in the original global coordinate system.

7. The anisotropic parabolic band

(a) Critical point and its type

The gradient and Hessian are

$$\nabla_{\boldsymbol{k}}\varepsilon = \left(\frac{2(k_x - k_0)}{\alpha}, \frac{2k_y}{\beta}, \frac{2k_z}{\gamma}\right), \qquad H = \mathrm{diag}(2/\alpha, 2/\beta, 2/\gamma).$$

All three Hessian eigenvalues are positive. Hence

$$\boxed{\boldsymbol{k}_c = (k_0, 0, 0), \qquad \varepsilon_c = \varepsilon_0, \qquad \text{type } M_0 \text{ (minimum)}}.$$

(b) Density of states

For $\Delta = E - \varepsilon_0 \geq 0$, the states below E occupy an ellipsoid with semiaxes $\sqrt{\alpha\Delta}$, $\sqrt{\beta\Delta}$, and $\sqrt{\gamma\Delta}$. Its volume is $(4\pi/3)\sqrt{\alpha\beta\gamma}\,\Delta^{3/2}$. There is one allowed wave vector per volume $(2\pi)^3/V$ in reciprocal space. Thus the integrated state count is

$$\mathcal{N}(E) = \frac{g_s V\sqrt{\alpha\beta\gamma}}{6\pi^2}(E - \varepsilon_0)^{3/2}\Theta(E - \varepsilon_0).$$

Differentiation gives

$$\boxed{D(E) = \frac{g_s V\sqrt{\alpha\beta\gamma}}{4\pi^2}\sqrt{E - \varepsilon_0}\,\Theta(E - \varepsilon_0)}.$$

Use $g_s = 1$ for a spin-resolved band, or $g_s = 2$ for a band whose two spin states are counted together. Translation of the ellipsoid by k_0 leaves its volume unchanged. If the parabola is only a local band-edge approximation, this expression applies near that edge, before other band features or zone boundaries matter.

8. Three valence electrons per atom in an fcc metal

An fcc conventional cell contains four atoms. The stated three valence electrons per atom therefore give

$$n = \frac{N}{V} = \frac{12}{a^3}.$$

By definition, $4\pi(r_s a_0)^3/3 = 1/n$, where a_0 is the Bohr radius. Hence

$$\boxed{r_s = \frac{a}{(16\pi)^{1/3} a_0}}.$$

Counting the two spin states in a filled Fermi sphere gives $n = k_F^3/(3\pi^2)$, so

$$\boxed{k_F = \frac{(36\pi^2)^{1/3}}{a}}, \qquad \boxed{\varepsilon_F = \frac{\hbar^2}{2m_e a^2}(36\pi^2)^{2/3}}.$$

Finally,

$$\overline{\varepsilon} = \frac{\int_0^{k_F} (\hbar^2 k^2/2m_e) k^2 \, \mathrm{d}k}{\int_0^{k_F} k^2 \, \mathrm{d}k} = \frac{\hbar^2}{2m_e} \frac{k_F^5/5}{k_F^3/3} = \boxed{\frac{3}{5}\varepsilon_F}.$$

This uses the stipulated free-electron model; it does not assume that actual multivalent-metal bands are exactly free-electron dispersions.

9. Sommerfeld expansion at fixed particle number

The expansion requires $T \ll T_F$ and a density of states smooth near the chemical potential. The density of states is zero below the bottom of the band, so the printed lower integration limit $-\infty$ is harmless.

(a) Temperature derivative of the chemical potential

Apply the supplied formula with $\Gamma = D$:

$$N = \int_{-\infty}^{\mu(T)} D(E) \, \mathrm{d}E + \frac{\pi^2}{6} k_\mathrm{B}^2 T^2 D'(\mu(T)) + O(T^4).$$

At fixed N and volume, differentiation gives

$$0 = D(\mu)\mu' + \frac{\pi^2}{3} k_\mathrm{B}^2 T D'(\mu) + \frac{\pi^2}{6} k_\mathrm{B}^2 T^2 D''(\mu)\mu' + O(T^3),$$

where $\mu' = \mathrm{d}\mu/\,\mathrm{d}T$. The leading change in μ is quadratic in T, so $\mu' = O(T)$ and the third term is $O(T^3)$. Keeping the requested order,

$$\boxed{\frac{\mathrm{d}\mu}{\mathrm{d}T} = -\frac{\pi^2}{3} k_\mathrm{B}^2 T \frac{D'(\mu)}{D(\mu)} + O(T^3)}.$$

The volume is fixed, so no temperature-dependent volume factor is differentiated.

(b) The three-dimensional free-electron result

For $E > 0$, $D(E) = C\sqrt{E}$ and $D'/D = 1/(2E)$. Therefore

$$\frac{\mathrm{d}\mu}{\mathrm{d}T} = -\frac{\pi^2 k_\mathrm{B}^2 T}{6\mu} + O(T^3).$$

Multiply by 2μ and integrate from 0 to T:

$$\mu(T)^2 = \mu_0^2 - \frac{\pi^2}{6} k_\mathrm{B}^2 T^2 + O(T^4).$$

Expanding the positive square root, and using $\mu_0 = \varepsilon_F = k_\mathrm{B} T_F$, yields

$$\boxed{\mu(T) = \mu_0 \left[1 - \frac{\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right] + O(T^4)}.$$

Here and below, the remainder notation in dimensional formulas carries the appropriate constant energy and temperature scales.

(c) Internal energy

Set $\Gamma(E) = ED(E)$. Then $\Gamma'(E) = D(E) + ED'(E)$, giving

$$\boxed{U(T) = \int_{-\infty}^{\mu(T)} ED(E)\,\mathrm{d}E + \frac{\pi^2}{6}(k_{\mathrm{B}}T)^2\big[D(\mu(T)) + \mu(T)D'(\mu(T))\big] + O(T^4)}.$$

The upper limit changes with temperature because the number of electrons is held fixed.

(d) Constant-volume electronic heat capacity

Differentiate the preceding expression along the fixed-N solution $\mu(T)$:

$$\begin{aligned} C_V &= \mu D(\mu)\mu' + \frac{\pi^2}{3}k_{\mathrm{B}}^2 T\big[D(\mu) + \mu D'(\mu)\big] + O(T^3) \\ &= -\frac{\pi^2}{3}k_{\mathrm{B}}^2 T\mu D'(\mu) + \frac{\pi^2}{3}k_{\mathrm{B}}^2 T\big[D(\mu) + \mu D'(\mu)\big] + O(T^3) \\ &= \frac{\pi^2}{3}k_{\mathrm{B}}^2 T D(\mu) + O(T^3). \end{aligned}$$

The two terms involving $\mu D'$ cancel. Since $\mu - \mu_0 = O(T^2)$, replacing $D(\mu)$ by $D(\mu_0)$ changes only the cubic correction:

$$\boxed{C_V(T) = \frac{\pi^2}{3}k_{\mathrm{B}}^2 T D(\mu_0) + O(T^3)}.$$

(e) Why the classical value is strongly suppressed

Deep occupied states cannot be thermally rearranged freely because of Pauli exclusion. Only a shell of width of order $k_{\mathrm{B}}T$ near ε_F contributes appreciably. In three dimensions,

$$D(\varepsilon_F) = \frac{3N}{2\varepsilon_F}, \qquad N_{\text{active}} \sim D(\varepsilon_F)k_{\mathrm{B}}T = \frac{3}{2}N\frac{T}{T_F}.$$

Estimating an order-k_{B} contribution per active state reproduces the printed scale $C_V \sim (3/2)Nk_{\mathrm{B}}(T/T_F)$ and explains why it is much smaller than the classical $(3/2)Nk_{\mathrm{B}}$.

The printed estimate does not have the exact coefficient. Substitution of the same $D(\varepsilon_F)$ into part (d) instead gives

$$\boxed{C_V = \frac{\pi^2}{2}Nk_{\mathrm{B}}\frac{T}{T_F} + O\big(Nk_{\mathrm{B}}(T/T_F)^3\big)}.$$

Thus part (e) is a qualitative active-electron argument. Treating its factor $3/2$ as exact would contradict part (d).

References and checks

The lattice, state-counting, and Sommerfeld derivations are displayed explicitly above. For the statistical-mechanical assumptions and low-temperature expansion, see David Tong, Statistical Physics, Chapter 3. The accompanying verification script independently counts finite lattice samples and integrates the finite-temperature Fermi distribution to check the limiting coefficients.

Chapter 8

Solid State Theory: Final, June 24, 2020

All twelve visible questions are addressed. The supplied page has a total weight of 85%. The derivations below distinguish an exchange operator from an exchange-correlation energy functional, and count bands per primitive cell rather than per conventional cubic cell.

1. Hartree and Hartree–Fock

Write $x = (\boldsymbol{r}, \sigma)$ for position and spin. A Hartree trial state is a product,

$$\Psi_{\mathrm{H}}(x_1, \ldots, x_N) = \prod_{i=1}^{N} \psi_i(x_i).$$

It describes each electron moving in an average electrostatic field. The product itself is not antisymmetric under exchange of two electrons and does not generate the Fock exchange term. Hartree–Fock (HF) instead uses a Slater determinant of orthonormal spin orbitals,

$$\Psi_{\mathrm{HF}} = \frac{1}{\sqrt{N!}} \det[\psi_i(x_j)].$$

The determinant changes sign upon exchange and therefore implements the Pauli principle. Let $\hat{h} = -\hbar^2\nabla^2/(2m) + v_{\mathrm{ext}}$ and define

$$J_{ij} = \iint |\psi_i(x)|^2 \frac{e^2}{|\boldsymbol{r} - \boldsymbol{r}'|} |\psi_j(x')|^2 \, \mathrm{d}x \, \mathrm{d}x',$$
$$K_{ij} = \iint \psi_i^*(x)\psi_j(x) \frac{e^2}{|\boldsymbol{r} - \boldsymbol{r}'|} \psi_j^*(x')\psi_i(x') \, \mathrm{d}x \, \mathrm{d}x'.$$

Then

$$E_{\mathrm{HF}} = \sum_i \langle \psi_i | \hat{h} | \psi_i \rangle + \frac{1}{2} \sum_{ij} (J_{ij} - K_{ij}).$$

For pure spin states, exchange acts only between equal spins. The terms with $i = j$ cancel because $J_{ii} = K_{ii}$; an electron does not interact with itself in HF. The Hartree product energy contains the direct interactions with $j \neq i$ but no exchange. A common density-only Hartree approximation includes self-interaction unless it is separately removed.

HF treats exchange exactly within a single determinant. It still misses correlation beyond that determinant. Its exchange operator is generally nonlocal; it must not be identified with the local Kohn–Sham exchange-correlation potential.

2. Fully spin-polarized homogeneous gas

Hold N and V fixed. In the normal, unpolarized state, two spin species occupy equal Fermi spheres, so

$$n = \frac{2}{(2\pi)^3}\frac{4\pi k_F^3}{3} = \frac{k_F^3}{3\pi^2}.$$

In the fully polarized state only one spin species is occupied:

$$n = \frac{(k_F^{\mathrm{FP}})^3}{6\pi^2}, \qquad k_F^{\mathrm{FP}} = 2^{1/3}k_F.$$

The kinetic energy per electron of a filled three-dimensional sphere is $3\hbar^2k_F^2/(10m)$. The exchange energy per electron of one occupied spin species is $-3e^2k_F/(4\pi)$. The latter has the same form in the normal state because each spin species exchanges only with itself. Therefore

$$\boxed{E_0^{(\mathrm{FP})} = N\left[2^{2/3}\frac{3}{5}\frac{\hbar^2k_F^2}{2m} - 2^{1/3}\frac{3e^2k_F}{4\pi}\right].}$$

The k_F in this box is the *normal-state* Fermi wavevector in the question. Reusing it as the polarized wavevector without the factor $2^{1/3}$ would violate particle-number conservation.

As a check, define the polarization $\zeta = (N_\uparrow - N_\downarrow)/N$. Since $k_{F\sigma} = k_F(1 \pm \zeta)^{1/3}$,

$$\frac{T(\zeta)}{N} = \frac{3}{5}E_F\frac{(1+\zeta)^{5/3} + (1-\zeta)^{5/3}}{2},$$
$$\frac{E_x(\zeta)}{N} = -\frac{3e^2k_F}{4\pi}\frac{(1+\zeta)^{4/3} + (1-\zeta)^{4/3}}{2}.$$

Setting $\zeta = 0$ and 1 recovers both states. Polarization raises kinetic energy and makes exchange more negative. It does not automatically lower their sum; the competition depends on density.

3. Equations and self-consistent solution

For individually normalized Hartree orbitals, variation of the product-state energy gives

$$\left[\hat{h} + \sum_{j\neq i} J_j(\boldsymbol{r})\right]\psi_i(x) = \epsilon_i\psi_i(x), \qquad J_j(\boldsymbol{r}) = \int \frac{e^2|\psi_j(x')|^2}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'.$$

Here $\hat{h}$ contains the kinetic energy and the fixed external potential, while J_j is the electrostatic repulsion from orbital j. The explicitly excluded i term avoids self-interaction in this product-state variational equation.

For HF, impose $\langle\psi_i \mid \psi_j\rangle = \delta_{ij}$ using a Hermitian matrix of Lagrange multipliers. Variation of

$$E_{\mathrm{HF}} - \sum_{ij}\lambda_{ij}(\langle\psi_i \mid \psi_j\rangle - \delta_{ij})$$

with respect to ψ_i^* yields a coupled orbital equation. A unitary rotation among occupied orbitals diagonalizes λ, giving the canonical form

$$\boxed{\left[\hat{h} + \sum_j(\hat{J}_j - \hat{K}_j)\right]\psi_i = \epsilon_i\psi_i.}$$

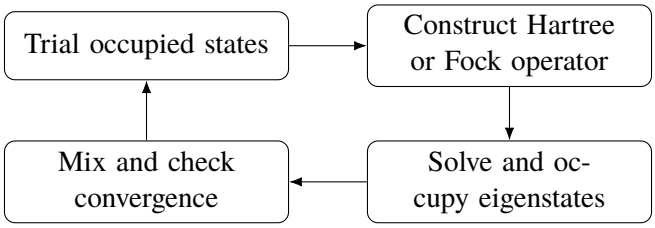
The exchange operator acts as

$$(\hat{K}_j\psi_i)(x) = \psi_j(x)\int \frac{e^2\psi_j^*(x')\psi_i(x')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}x'.$$

It depends on ψ_i at other positions and is not multiplication by a universal local scalar. Including all j is safe because the direct and exchange self-terms cancel.

Both problems are nonlinear: their operators depend on the unknown occupied states. A self-consistent-field iteration is:

1. Choose trial orbitals or a trial density matrix and the desired occupations.
2. Construct the direct field and, for HF, the exchange operator.
3. Solve the resulting one-electron eigenproblem and occupy the required states.
4. Mix old and new density matrices or orbitals if needed; repeat until the energy and density or commutator residual converge.
5. Check the particle number, orbital normalization, and possible competing spin or symmetry solutions. A converged stationary solution need not be the lowest-energy one.



4. What DFT adds

Ground-state density functional theory (DFT) makes the three-dimensional density the basic variable. Its exact universal functional contains both exchange and correlation, whereas ordinary HF restricts the state to a single determinant and omits the remaining correlation energy. Practical Kohn–Sham DFT introduces auxiliary orbitals to evaluate the noninteracting kinetic energy. With a local or semilocal exchange-correlation approximation, it avoids the full nonlocal Fock exchange operation and is often less expensive.

These are formal and computational advantages, not a guarantee that every approximate density functional is more accurate than HF. The exact functional is not generally available; approximate functionals can have self-interaction, band-gap, dispersion, or strong-correlation errors. Kohn–Sham DFT still solves for occupied orbitals and is not simply one scalar differential equation for the density.

5. Hohenberg–Kohn energy functional

For fixed interaction and particle number, a precise universal functional can be written using a constrained search:

$$F[n] = \min_{\Psi\to n}\langle\Psi|\hat{T}+\hat{W}|\Psi\rangle.$$

The minimization is over admissible antisymmetric many-electron states with density n. With an external potential v_{ext}, the energy functional is

$$\boxed{E_v[n] = F[n] + \int v_{\text{ext}}(\boldsymbol{r})n(\boldsymbol{r})\,\mathrm{d}^3r.}$$

A fixed ion–ion energy may be added as a constant. The universal $F[n]$ does not explicitly depend on v_{ext}; the full $E_v[n]$ does. Among admissible densities with $\int n = N$, the ground-state density minimizes $E_v[n]$ and gives the ground-state energy. Different external potentials normally produce different minimizing densities.

In the original nondegenerate Hohenberg–Kohn setting, the ground-state density determines the external potential up to an additive constant. The constrained-search formulation expresses the energy minimization without needing to assign every trial density to a unique nondegenerate external-potential ground state.

6. Kohn–Sham equations

Decompose the universal functional as

$$F[n] = T_s[n] + E_{\text{H}}[n] + E_{\text{xc}}[n], \qquad E_{\text{H}}[n] = \frac{1}{2}\iint \frac{e^2 n(\boldsymbol{r}) n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3 r\,\mathrm{d}^3 r'.$$

Here T_s is the kinetic energy of an auxiliary noninteracting system with density n. The definition

$$E_{\text{xc}}[n] = F[n] - T_s[n] - E_{\text{H}}[n]$$

includes exchange, the interaction correlation beyond the Hartree term, and the difference between interacting and noninteracting kinetic energies.

For a spin-independent potential, the equations are

$$\left[-\frac{\hbar^2\nabla^2}{2m} + v_{\text{ext}}(\boldsymbol{r}) + v_{\text{H}}(\boldsymbol{r}) + v_{\text{xc}}(\boldsymbol{r})\right]\phi_i(\boldsymbol{r}) = \epsilon_i \phi_i(\boldsymbol{r}),$$

$$n(\boldsymbol{r}) = \sum_i f_i |\phi_i(\boldsymbol{r})|^2,$$

$$v_{\text{H}}(\boldsymbol{r}) = \int \frac{e^2 n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3 r', \qquad v_{\text{xc}}(\boldsymbol{r}) = \frac{\delta E_{\text{xc}}}{\delta n(\boldsymbol{r})}.$$

The occupations satisfy $\sum_i f_i = N$. They are at most two for spatial orbitals in a spin-degenerate treatment, or at most one for explicitly labelled spin orbitals. Spin-polarized calculations use $E_{\text{xc}}[n_\uparrow, n_\downarrow]$ and a potential for each spin. Self-consistency is required because the potential is a functional of the density obtained from the orbitals.

7. Deriving Kohn–Sham from the energy functional

Represent the density by orthonormal auxiliary orbitals and write

$$T_s = -\sum_i f_i \int \phi_i^* \frac{\hbar^2\nabla^2}{2m}\phi_i\,\mathrm{d}^3 r.$$

Vary $E_v[n]$ with respect to ϕ_i^* while constraining orbital orthonormality. Since $\delta n = f_i \phi_i \delta\phi_i^*$ for this variation,

$$\frac{\delta T_s}{\delta \phi_i^*} = -f_i \frac{\hbar^2\nabla^2}{2m}\phi_i,$$

$$\frac{\delta(E_{\text{H}} + E_{\text{xc}} + \int v n)}{\delta \phi_i^*} = f_i(v_{\text{H}} + v_{\text{xc}} + v)\phi_i.$$

The Euler–Lagrange equations are consequently the Kohn–Sham equations of Question 6, after diagonalizing the constraint matrix within equal-occupation subspaces. They are a way to minimize the density functional while treating T_s through orbitals; the orbitals are not generally the exact interacting many-electron wavefunction.

An independent consistency check is the total-energy expression. Summing occupied Kohn–Sham eigenvalues counts the Hartree term twice and includes $\int n v_{\mathrm{xc}}$, rather than E_{xc}. Thus

$$\boxed{E = \sum_i f_i \epsilon_i - E_{\mathrm{H}} - \int n v_{\mathrm{xc}}\, \mathrm{d}^3 r + E_{\mathrm{xc}} + E_{\mathrm{ion-ion}}.}$$

The sum of eigenvalues alone is not the total energy.

8. Functional derivative in the LDA

Because $\varepsilon_{\mathrm{xc}}(n)$ is an energy *per electron*, the LDA functional is

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int n(\boldsymbol{r}) \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r}))\, \mathrm{d}^3 r.$$

Under an arbitrary small density change,

$$\delta E_{\mathrm{xc}}^{\mathrm{LDA}} = \int \left[\varepsilon_{\mathrm{xc}}(n) + n \frac{\mathrm{d}\varepsilon_{\mathrm{xc}}}{\mathrm{d}n} \right] \delta n\, \mathrm{d}^3 r.$$

Reading off the coefficient of δn gives

$$\boxed{v_{\mathrm{xc}}^{\mathrm{LDA}}(\boldsymbol{r}) = \left[\varepsilon_{\mathrm{xc}}(n) + n \varepsilon_{\mathrm{xc}}'(n) \right]\big|_{n=n(\boldsymbol{r})}.}$$

It is not merely $\varepsilon_{\mathrm{xc}}$, nor merely its derivative. For example, if $\varepsilon_x = -Cn^{1/3}$, then $v_x = -(4C/3)n^{1/3}$. Consequently $\int n v_x = (4/3) E_x$, illustrating why the reference shortcut $E_{\mathrm{xc}} = \int n v_{\mathrm{xc}}$ is generally false.

9. Tight-binding Bloch sums

Let $\boldsymbol{R}$ run over N_c Bravais lattice sites, $\boldsymbol{\tau}_\alpha$ identify an atom in the primitive-cell basis, and $\varphi_{\alpha\mu}$ be a localized orbital. One convenient phase convention is

$$\chi_{\alpha\mu,\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N_c}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_\alpha)} \varphi_{\alpha\mu}(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_\alpha).$$

Translate by a Bravais vector $\boldsymbol{T}$ and substitute $\boldsymbol{R}' = \boldsymbol{R} - \boldsymbol{T}$. Then

$$\chi_{\alpha\mu,\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{T}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{T}} \chi_{\alpha\mu,\boldsymbol{k}}(\boldsymbol{r}),$$

so each sum has the Bloch transformation law. An eigenstate is a linear combination of these sums at fixed $\boldsymbol{k}$:

$$\psi_{n\boldsymbol{k}} = \sum_{\alpha\mu} c_{\alpha\mu,n}(\boldsymbol{k}) \chi_{\alpha\mu,\boldsymbol{k}}.$$

Projection gives $H(\boldsymbol{k}) c_n = E_n(\boldsymbol{k}) S(\boldsymbol{k}) c_n$. Here S is the orbital-overlap matrix. The displayed $1/\sqrt{N_c}$ yields individually normalized Bloch sums only when the localized basis is orthonormal, or its overlap is neglected; otherwise S or explicit normalization must be retained. The hopping matrix elements, not dimensionless overlaps alone, determine the energy dispersion.

10. Silicon orbitals and band counting

(a) Five orbitals per atom. Silicon has four chemically active valence orbitals, $3s$ and the three $3p$ orbitals. A common five-orbital empirical choice is

$$\boxed{\{s, p_x, p_y, p_z, s^*\}.}$$

The extra excited s-like orbital improves the description of the low conduction bands. It is an effective basis state, not a fifth occupied valence orbital in the isolated ground-state atom. Core $1s$, $2s$, and $2p$ states are normally excluded from this valence-band model.

(b) Number of bands. Diamond-structure silicon has two atoms per primitive cell. Five spatial orbitals on each atom produce a 10×10 Bloch Hamiltonian, hence $\boxed{10 \text{ spatial bands}}$. With spin degeneracy, each band accommodates two electrons per primitive cell. Counting spin-resolved branches instead gives 20 coincident branches when spin splitting is absent. Using the eight atoms of the conventional cubic cell would fold the bands and change the bookkeeping, not create new physical states.

(c) Filled bands. There are $2 \times 4 = 8$ valence electrons per primitive cell. In the insulating ground state, each full spin-degenerate band holds two, so

$$\boxed{N_{\text{filled}} = \frac{8}{2} = 4.}$$

The remaining six spatial bands in the model are unoccupied. Counting only one atom would incorrectly give two occupied bands.

11. Why bare plane waves converge slowly near cores

A plane wave is smooth, but an all-electron valence state has rapid variations near nuclei and must be orthogonal to compact core states. Resolving these short length scales requires large reciprocal vectors $\boldsymbol{G}$ and a high kinetic-energy cutoff,

$$\frac{\hbar^2 |\boldsymbol{k} + \boldsymbol{G}|^2}{2m} \leq E_{\text{cut}}.$$

In three dimensions the number of waves grows approximately as $E_{\text{cut}}^{3/2}$. A finite plane-wave variational calculation that does not constrain the valence subspace can also approach lower core states. This is a failure to isolate the intended valence problem, not an unbounded-below nonrelativistic Hamiltonian.

Three related remedies are useful:

1. **Orthogonalized plane waves (OPW).** Project every plane wave out of the occupied core space: $(1 - P_c)|\boldsymbol{k} + \boldsymbol{G}\rangle$. Core orthogonality is then built into the basis.

2. **Pseudopotentials.** Replace the strong core potential and nodal all-electron valence function by an effective potential and a smooth pseudo-wavefunction that reproduce the relevant valence scattering and external behavior. Fewer plane waves are needed.

3. **Augmentation methods.** Use an appropriate near-nucleus representation together with a smooth interstitial representation, as in augmented-wave approaches.

If $|\psi\rangle = (1 - P_c)|\phi\rangle$ and $H|\chi_c\rangle = \epsilon_c|\chi_c\rangle$, the OPW construction implies

$$\left[H + \sum_c (E - \epsilon_c) |\chi_c\rangle\langle\chi_c| \right] |\phi\rangle = E|\phi\rangle.$$

For a valence state above the core energies, the additional projector term is repulsive. This is the Phillips–Kleinman connection between core orthogonality and an effective pseudopotential.

12. Core Bloch sums for LiH

In the usual rocksalt description, a primitive cell contains one Li and one H. The standard valence partition freezes the tightly bound Li $1s^2$ shell as core; Li $2s$ and H $1s$ contribute to the valence sector. Although the occupied valence states have substantial H character, H $1s$ is not an additional frozen core in this partition.

For Li sites $\boldsymbol{R} + \boldsymbol{\tau}_{\mathrm{Li}}$, the spatial core Bloch sum is

$$\boxed{\chi_{c\boldsymbol{k}}(\boldsymbol{r}) = \mathcal{N}_{\boldsymbol{k}} \sum_{\boldsymbol{R}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{\tau}_{\mathrm{Li}})} \varphi_{\mathrm{Li},1s}(\boldsymbol{r} - \boldsymbol{R} - \boldsymbol{\tau}_{\mathrm{Li}}).}$$

There is one such spatial core band, doubly occupied when spin is included. If intersite core overlap is negligible, $\mathcal{N}_{\boldsymbol{k}} = 1/\sqrt{N_c}$; otherwise normalize the sum or orthonormalize the core subspace. For each spin, the OPW is

$$|w_{\boldsymbol{k}+\boldsymbol{G}}\rangle = |\boldsymbol{k}+\boldsymbol{G}\rangle - |\chi_{c\boldsymbol{k}}\rangle\langle\chi_{c\boldsymbol{k}} \mid \boldsymbol{k}+\boldsymbol{G}\rangle.$$

Only a core Bloch sum at the same crystal momentum enters because different Bloch sectors are orthogonal. A different chosen frozen-core partition must be specified explicitly before changing the projector.

Primary references and checks

The algebra is derived above. The following original papers establish the functional and basis constructions used for comparison:

- P. Hohenberg and W. Kohn, *Inhomogeneous Electron Gas* (1964).
- W. Kohn and L. J. Sham, *Self-Consistent Equations Including Exchange and Correlation Effects* (1965).
- J. C. Phillips and L. Kleinman, *New Method for Calculating Wave Functions in Crystals and Molecules* (1959).
- P. Vogl, H. P. Hjalmarson, and J. D. Dow, *A Semi-empirical Tight-binding Theory of the Electronic Structure of Semiconductors* (1983).

Chapter 9

Solid State Theory: April 23, 2021 — Solutions

These are independent solutions, not an official answer key. Several true/false statements require a convention or a qualification; those conditions are stated rather than hidden. Reciprocal vectors use $\mathbf{g}_i \cdot \mathbf{t}_j = 2\pi\delta_{ij}$.

1. True/false, with reasons and necessary qualifications

(a) Diamond sites form a Bravais lattice: false

Diamond consists of an FCC Bravais lattice with a two-atom basis, for example $\mathbf{d}_1 = 0, \mathbf{d}_2 = (a/4)(1,1,1)$. Translating by $\mathbf{d}_2$ sends the first atom to the second but does not map the entire crystal to itself: twice that displacement is not a diamond site relative to the chosen origin. The carbon sites are therefore not a one-site Bravais lattice.

(b) The two vector pairs span the same lattice: false

The primitive parallelogram areas would be

$$|\det(\mathbf{t}_1, \mathbf{t}_2)| = |2ab - ab| = ab, \qquad |\det(\mathbf{s}_1, \mathbf{s}_2)| = |-ab + 6ab| = 5ab.$$

More explicitly,

$$\mathbf{s}_1 = 4\mathbf{t}_1 + 7\mathbf{t}_2, \qquad \mathbf{s}_2 = -3\mathbf{t}_1 - 4\mathbf{t}_2, \quad M = \begin{pmatrix} 4 & -3 \\ 7 & -4 \end{pmatrix}, \quad \det M = 5.$$

An integer basis change preserves a lattice only if its determinant is ± 1. Here the $\mathbf{s}$ vectors generate an index-5 sublattice of the $\mathbf{t}$ lattice, with only one fifth as many sites per unit area.

(c) Same point group implies same crystal system: true

With point group understood as the full crystallographic point group, two Bravais lattices with that same group belong to the same crystal system. They may differ in centering. For example, primitive, body-centered, and face-centered cubic Bravais lattices are distinct but all belong to the cubic system. Sharing merely a subgroup, such as inversion alone, would not suffice.

(d) Same species implies a Bravais translation: false in general

Positions have the form $\mathbf{R} + \mathbf{d}_\alpha$. The difference of two positions is a Bravais vector only if their basis-site difference is also a Bravais vector. Diamond supplies two carbon atoms whose separation $(a/4)(1,1,1)$ is not an FCC translation. The phrase *simple crystal* is not defined in the question. If it is explicitly defined in a course as a one-atom Bravais crystal, the statement is true by that extra definition. Chemical purity or a perfect defect-free crystal alone does not establish it.

(e) Wigner-Seitz and Brillouin-zone product: true for volumes

The meaningful interpretation is the product of their volumes. Write the direct and reciprocal basis matrices as T and $G = 2\pi T^{-T}$. Then

$$\Omega = |\det T|, \qquad \Omega^* = |\det G| = \frac{(2\pi)^3}{\Omega}, \qquad \boxed{\Omega\Omega^* = (2\pi)^3.}$$

Wigner-Seitz cells have the same volume as the corresponding primitive parallelepipeds. The constant depends on the reciprocal-vector convention; it would be 1 if the 2π were omitted throughout.

(f) Allowed k-point density scales with volume: true

For N_i cells along each primitive direction, Born-von Karman conditions give

$$\mathbf{k} = \sum_i \frac{m_i}{N_i}\mathbf{g}_i, \qquad m_i = 0, \ldots, N_i - 1.$$

There are $N_c = N_1 N_2 N_3$ inequivalent points per Brillouin zone. If the sample volume is $V = N_c\Omega$, each point occupies reciprocal volume $(2\pi)^3/V$ and

$$\boxed{\frac{\mathrm{d}N_k}{\mathrm{d}^3k} = \frac{V}{(2\pi)^3}.}$$

This counts wavevectors; each band and spin supplies additional electronic states. Increasing the sample volume makes the k mesh denser, not sparser.

(g) One cell determines a Bloch eigenstate elsewhere: true

For a chosen Bloch state,

$$\psi_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}}\psi_{n\mathbf{k}}(\mathbf{r}).$$

Knowing the wavefunction within one cell and its $\mathbf{k}$ therefore determines it in every translated cell. The qualification is that a Bloch eigenbasis is chosen: arbitrary linear combinations of degenerate states with different $\mathbf{k}$ need not possess a single translation phase.

(h) HF energy must be strictly lower than Hartree: qualified

As an unconditional assertion of a *strict* inequality, the statement is false. The standard negative-exchange argument compares a fixed orthonormal orbital set:

$$E_{\text{HF}} - E_{\text{Hartree product}} = -\frac{1}{2}\sum_{i\neq j} K_{ij} \leq 0,$$

Here $x = (\mathbf{r}, \sigma)$ includes spin, integration over x includes the spin sum, and $w(\mathbf{r} - \mathbf{r}') = e^2/(4\pi\varepsilon_0|\mathbf{r} - \mathbf{r}'|)$. The exchange integral is

$$K_{ij} = \int \mathrm{d}x\, \mathrm{d}x'\, \chi_i^*(x)\chi_j(x)w(\mathbf{r} - \mathbf{r}')\chi_j^*(x')\chi_i(x') \geq 0.$$

Thus exchange lowers or leaves unchanged that reference energy. Opposite-spin pairs have zero exchange, and equality is possible when all relevant off-diagonal exchange integrals vanish. A strict decrease needs at least one nonzero such integral. Separately optimized Hartree products and HF determinants also have different variational constraints; a universal ordering cannot be inferred simply by calling one wavefunction antisymmetric. Under the usual classroom fixed-orbital comparison with nonzero exchange, the intended lower-energy conclusion holds with these qualifications.

(i) A self-consistent HF determinant is an exact eigenfunction: false

HF makes the energy stationary under variations within the single-determinant manifold. The interacting Hamiltonian can couple that determinant to excited determinants, especially double excitations. In general, $\hat{H}\Phi_{\mathrm{HF}} \neq E\Phi_{\mathrm{HF}}$. An exact single-determinant eigenstate is a special case, not a consequence of solving the HF equations.

(j) A self-consistent HF determinant is the best many-body state: false as written

The variational minimum satisfies

$$E_0 \leq \min_{\Phi \text{ single determinant}} \langle\Phi|\hat{H}|\Phi\rangle = E_{\mathrm{HF}}^{\mathrm{min}}.$$

Correlated wavefunctions outside this restricted class can lower the energy. Moreover, a solution of the HF equations can be a local minimum or another stationary point, rather than the global determinant minimum. The precise valid claim is: *the globally minimizing HF determinant is the best energy approximation within the class of allowed single determinants*. It need not be the best wavefunction among all many-electron states.

(k) The unpolarized homogeneous gas always has lower energy: false

Interpret the normal state as the unpolarized uniform state. The competition depends on density: polarization increases Fermi kinetic energy but makes exchange more negative. This is explicit in the uniform Hartree-Fock model. In atomic units, let

$$k_F = (3\pi^2 n)^{1/3} = \frac{c_0}{r_s}, \quad c_0 = (9\pi/4)^{1/3}, \quad \zeta = \frac{n_\uparrow - n_\downarrow}{n}.$$

With the positive background canceling the uniform Hartree term, the energy per electron is

$$e_{\mathrm{HF}}(\zeta) = \frac{3k_F^2}{20}\left[(1+\zeta)^{5/3} + (1-\zeta)^{5/3}\right] - \frac{3k_F}{8\pi}\left[(1+\zeta)^{4/3} + (1-\zeta)^{4/3}\right].$$

The powers follow from each spin Fermi sphere: kinetic energy density scales as $n_\sigma^{5/3}$ and exchange as $-n_\sigma^{4/3}$. Comparing $\zeta = 1$ with $\zeta = 0$ gives

$$\Delta e = \frac{3k_F^2}{10}(2^{2/3} - 1) - \frac{3k_F}{4\pi}(2^{1/3} - 1).$$

At high density the first term wins; at lower density the second can win. These two uniform HF branches cross at

$$r_s^* = \frac{2\pi c_0}{5}(2^{1/3} + 1) \simeq 5.45.$$

This example concerns uniform HF states; it is not a claim that this is the exact phase boundary of the correlated electron gas. Correlation and other possible ordered states require separate treatment. There is no density-independent energy-ordering theorem of the asserted form.

(l) Electron density depends on external potential: true for the physical density

For a fixed electron number and interaction, the physical density is the minimizer

$$n_0[v] = \arg\min_n \left(F[n] + \int v(\mathbf{r}) n(\mathbf{r}) \, \mathrm{d}^3 r \right).$$

Changing v generally changes n_0. What is universal is $F[n]$: it has no *explicit* dependence on the external potential. A trial density used in the variation is an independent argument. Confusing that functional argument with the resulting minimizing density leads to the wrong conclusion.

(m) KS orbitals form the interacting ground-state determinant: false

KS orbitals define an auxiliary noninteracting state chosen to reproduce the interacting ground-state *density*. Its Slater determinant is generally not the interacting many-electron wavefunction. Matching a one-body density does not reproduce all pair correlations or all expectation values.

(n) Both density and interacting kinetic energy are KS sums: false

With exact KS orbitals and the appropriate occupations,

$$n_0(\mathbf{r}) = \sum_i f_i |\phi_i(\mathbf{r})|^2$$

is correct. But

$$T_s = \sum_i f_i \left\langle \phi_i \left| \frac{\mathbf{p}^2}{2m} \right| \phi_i \right\rangle$$

is the auxiliary noninteracting kinetic energy, not in general the interacting T. The kinetic correlation contribution $T - T_s$ is part of E_{xc}. A conjunction is false when its second assertion fails.

(o) KS gives the exact density and energy: true only in the exact theory

With the exact exchange-correlation functional, an appropriate KS representation, and the ground-state self-consistent solution, the formal theory yields the exact density and ground-state electronic energy:

$$E_0 = T_s[n_0] + E_H[n_0] + E_{xc}[n_0] + \int v_{\text{ext}} n_0 \, \mathrm{d}^3 r.$$

This does not make practical LDA or other approximate functionals exact. Nor does it identify every KS eigenvalue with a physical excitation energy. The intended formal-theory answer is true; the unconditional assertion about all numerical KS calculations would be false.

2. Hamiltonian and density-functional terminology

(a) Fixed-nuclei many-electron Hamiltonian

For nuclei of charge $Z_A e$ at fixed positions $\mathbf{R}_A$,

$$\hat{H}_e = \sum_{i=1}^{N} \left[-\frac{\hbar^2}{2m}\nabla_i^2 - \sum_A \frac{Z_A e^2}{4\pi\varepsilon_0 |\mathbf{r}_i - \mathbf{R}_A|} \right] + \sum_{i<j} \frac{e^2}{4\pi\varepsilon_0 |\mathbf{r}_i - \mathbf{r}_j|},$$

$$E_{NN} = \sum_{A<B} \frac{Z_A Z_B e^2}{4\pi\varepsilon_0 |\mathbf{R}_A - \mathbf{R}_B|}.$$

The three electronic terms are kinetic energy, electron-nucleus attraction, and electron-electron repulsion. The full fixed-configuration energy includes the constant E_{NN}. Nuclear kinetic energy is absent because the nuclei are held fixed. The $i < j$ and $A < B$ restrictions count each pair once. Other applied external potentials can be added to the one-electron potential. The electronic wavefunction must be antisymmetric in electron space-spin coordinates.

(b) Hohenberg-Kohn functional

The universal interacting functional can be expressed by constrained search:

$$F[n] = \min_{\Psi \to n} \langle \Psi | \hat{T} + \hat{W}_{ee} | \Psi \rangle.$$

For a specified external potential,

$$E_v[n] = F[n] + \int v(\mathbf{r}) n(\mathbf{r}) \, \mathrm{d}^3 r, \qquad E_0 = \min_{n \; madmissible, \int n = N} E_v[n].$$

The first HK theorem relates ground-state density to external potential, up to an additive constant under its usual assumptions; the second establishes the density variational principle. F is independent of which fixed nuclear configuration produces v, while its minimizing density is not.

(c) Exchange-correlation functional

Define the noninteracting kinetic functional $T_s[n]$ and classical Coulomb functional

$$E_H[n] = \frac{1}{2} \int \mathrm{d}^3 r \, \mathrm{d}^3 r' \, \frac{e^2 n(\mathbf{r}) n(\mathbf{r}')}{4\pi\varepsilon_0 |\mathbf{r} - \mathbf{r}'|}.$$

Then

$$\boxed{E_{xc}[n] = F[n] - T_s[n] - E_H[n] = (T[n] - T_s[n]) + (W_{ee}[n] - E_H[n]).}$$

The second expression refers to the constrained-search minimizing interacting state. Exchange, correlation beyond classical Coulomb repulsion, self-interaction cancellation, and the kinetic correction are included. The potential entering the KS equation is $v_{xc}(\mathbf{r}) = \delta E_{xc} / \delta n(\mathbf{r})$.

(d) Local density approximation

LDA uses the exchange-correlation energy per electron of a uniform gas at the local density:

$$\boxed{E_{xc}^{\mathrm{LDA}}[n] = \int n(\mathbf{r}) \varepsilon_{xc}^{\mathrm{unif}}(n(\mathbf{r})) \, \mathrm{d}^3 r.}$$

Its potential is $v_{xc}^{\mathrm{LDA}} = \varepsilon_{xc}^{\mathrm{unif}} + n \, \mathrm{d}\varepsilon_{xc}^{\mathrm{unif}} / \mathrm{d}n$, evaluated at the local n. It is local in the density and omits explicit gradients and nonlocal correlations. It is motivated by a locally uniform or slowly varying density, rather than by an assumption that all real crystals have uniform charge density.

3. FCC lattice and the plane perpendicular to a body diagonal

(a) Primitive volume and lattice-point density

A conventional cubic cell contains $8(1/8) + 6(1/2) = 4$ FCC lattice points, so

$$\boxed{\Omega = \frac{a^3}{4}, \qquad \rho_{\text{lattice}} = \frac{4}{a^3}.}$$

This is a lattice-point density; a separate multi-atom basis would change atom density, not the Bravais-point density.

(b) Direct and reciprocal primitive vectors

Choose

$$\mathbf{t}_1 = \frac{a}{2}(0,1,1), \quad \mathbf{t}_2 = \frac{a}{2}(1,0,1), \quad \mathbf{t}_3 = \frac{a}{2}(1,1,0).$$

Using $\mathbf{g}_1 = 2\pi(\mathbf{t}_2 \times \mathbf{t}_3)/\Omega$ and cyclic permutations gives

$$\boxed{\mathbf{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \mathbf{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \mathbf{g}_3 = \frac{2\pi}{a}(1,1,-1).}$$

Their dot products with the $\mathbf{t}_j$ verify the reciprocal relation. The reciprocal lattice is BCC, with conventional cubic lattice constant $4\pi/a$.

(c) A (111) layer is a triangular lattice

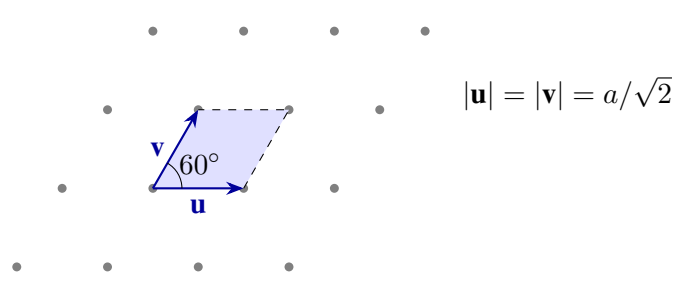
Take the normal along $(1,1,1)$. A basis in a layer is

$$\mathbf{u} = \mathbf{t}_2 - \mathbf{t}_1 = \frac{a}{2}(1,-1,0), \qquad \mathbf{v} = \mathbf{t}_3 - \mathbf{t}_1 = \frac{a}{2}(1,0,-1).$$

Both dot products with $(1,1,1)$ vanish. Their lengths are $a/\sqrt{2}$ and $\mathbf{u} \cdot \mathbf{v} = a^2/4$, hence their angle is $60°$. They span a two-dimensional triangular, or two-dimensional hexagonal Bravais, lattice. Its primitive area and planar density are

$$\boxed{A_{111} = \frac{\sqrt{3}a^2}{4}, \qquad \sigma_{111} = \frac{4}{\sqrt{3}a^2}.}$$

The shortest reciprocal normal is $(2\pi/a)(1,1,1)$, so adjacent lattice layers are separated by $a/\sqrt{3}$; indeed $A_{111}(a/\sqrt{3}) = a^3/4$. The triangular lattice of one plane should not be confused with the three-dimensional hexagonal lattice. Consecutive FCC (111) layers have the familiar ABC offsets.



4. Honeycomb reciprocal lattice, critical points, and DOS

(a) Reciprocal primitive vectors

The direct-cell area is $A_c = \sqrt{3}a^2/2$. Solving $\mathbf{g}_i \cdot \mathbf{t}_j = 2\pi\delta_{ij}$ gives

$$\boxed{\mathbf{g}_1 = \frac{2\pi}{a}(1, 1/\sqrt{3}), \qquad \mathbf{g}_2 = \frac{2\pi}{a}(-1, 1/\sqrt{3}).}$$

For example, the two equations for $\mathbf{g}_1$ are $ag_{1x}/2 + \sqrt{3}ag_{1y}/2 = 2\pi$ and $-ag_{1x}/2 + \sqrt{3}ag_{1y}/2 = 0$. The reciprocal primitive area is $8\pi^2/(\sqrt{3}a^2)$, and its product with A_c is $(2\pi)^2$.

(b) Classifying Γ, Q, and P

Set $x = ak_x$, $y = ak_y$, and $h = |F|^2$. Direct differentiation yields

$$h_x = -2\sin(x/2)\cos(\sqrt{3}y/2) - 4\cos(x/2)\sin(x/2), \qquad h_y = -2\sqrt{3}\cos(x/2)\sin(\sqrt{3}y/2).$$

For $h > 0$, a band is stationary exactly when $\nabla h = 0$.

At Γ, $h = 9$ and, for displacement $\mathbf{q}$,

$$|F| = 3 - \frac{a^2q^2}{4} + O(a^4q^4), \qquad \epsilon_\pm = \epsilon_p \pm 3\gamma \mp \frac{\gamma a^2 q^2}{4} + O(q^4).$$

Thus the upper band has a smooth maximum and the lower band a smooth minimum, with Hessians $\mp(\gamma a^2/2)I$.

At Q, $h = 1$ and the dimensionless Hessian is

$$H_h(Q) = \begin{pmatrix} 2 & \sqrt{3} \\ \sqrt{3} & 0 \end{pmatrix}.$$

Its eigenvectors can be chosen as $(\sqrt{3}/2, 1/2)$ and $(-1/2, \sqrt{3}/2)$, with eigenvalues $3, -1$. In their displacement coordinates u, v,

$$|F| = 1 + \frac{a^2}{4}(3u^2 - v^2) + O(a^3q^3), \qquad \epsilon_\pm = \epsilon_p \pm \gamma \pm \frac{\gamma a^2}{4}(3u^2 - v^2) + O(a^3q^3).$$

Each band has a saddle. In the original coordinates the quadratic correction to $|F|$ is $a^2q_x^2/2 + \sqrt{3}a^2q_xq_y/2$, so dropping the mixed term would give an incorrect classification.

At P, $h = 0$, and

$$h = \frac{3a^2q^2}{4} + O(a^3q^3), \qquad \epsilon_\pm = \epsilon_p \pm \frac{\sqrt{3}\gamma a}{2}|\mathbf{q}| + O(q^2).$$

This is a Dirac contact: a conical minimum in the upper band and a conical maximum in the lower band. The individual energy-ordered bands are not differentiable at the vertex, so there is no ordinary stationary gradient or quadratic Hessian there. Under a broader definition it is a band-contact critical singularity, not a smooth Morse critical point. Its one-sided directional slopes are finite.

(c) Choosing Q: derivation of the logarithmic DOS

For the upper band near one Q, define

$$E_Q = \epsilon_p + \gamma, \quad e = E - E_Q, \quad \alpha = \frac{3\gamma a^2}{4}, \quad \beta = \frac{\gamma a^2}{4}, \quad \epsilon - E_Q \simeq \alpha u^2 - \beta v^2.$$

The DOS per unit sample area per spin from that patch is

$$g_Q(E) = \frac{1}{(2\pi)^2} \int_{\text{patch}} \mathrm{d}u\, \mathrm{d}v\, \delta(e - \alpha u^2 + \beta v^2).$$

Let $X = \sqrt{\alpha}u$, $Y = \sqrt{\beta}v$, then $U = X + Y$, $V = X - Y$. The energy becomes UV and $\mathrm{d}u\, \mathrm{d}v = \mathrm{d}U\, \mathrm{d}V/(2\sqrt{\alpha\beta})$. Choose a small cutoff patch $|U|, |V| < L$. For $0 < |e| < L^2$, integration over V gives $1/|U|$ only when $|U| > |e|/L$. Hence

$$\begin{aligned} g_Q(E) &\simeq \frac{1}{8\pi^2\sqrt{\alpha\beta}} \int_{|e|/L<|U|<L} \frac{\mathrm{d}U}{|U|} \\ &= \frac{1}{4\pi^2\sqrt{\alpha\beta}} \log \frac{L^2}{|e|} \\ &= \boxed{\frac{1}{\pi^2\sqrt{3}\gamma a^2} \log \frac{E_c}{|E - E_Q|}, \qquad E_c = L^2.} \end{aligned}$$

The coefficient of the logarithm is fixed by the local curvatures. The cutoff scale and nonsingular background depend on the chosen patch and higher-order terms. The lower-band saddle has the same singular coefficient about $E_Q = \epsilon_p - \gamma$.

This counts a single saddle patch and one spin. Multiply by spin or distinct-saddle multiplicities only when summing those contributions, and multiply by A_c for DOS per cell. A patch centered on a zone edge is understood on the reciprocal-space torus, or split and translated into the first zone; it must not be counted twice. The logarithmic saddle singularity differs from the constant step at Γ and from the linear, vanishing DOS at a Dirac point.

Primary references and verification scope

The calculations of reciprocal vectors, Hessians, and local DOS are derived from the question's formulas. For the model and density-functional foundations, see Wallace, The Band Theory of Graphite (1947), Hohenberg and Kohn, Inhomogeneous Electron Gas (1964), and Kohn and Sham, Self-Consistent Equations Including Exchange and Correlation Effects (1965). The polarized-gas comparison is explicitly a uniform Hartree-Fock calculation and is not presented as an exact correlated phase diagram.

Chapter 10

Solid State Theory: Undated Midterm Fragment

Question 2 lacks its defining model. Its numerical answers cannot be recovered uniquely. The conditional method below is an additional study aid, not a claim about the missing original problem. Questions 3–7 can be answered from the surviving text. All band curvatures and density-of-states factors below are derived from the printed dispersion, independently of the handwritten references.

2. What can and cannot be inferred

Band gaps and effective masses depend on the Hamiltonian. Without the potential or the missing model parameters, there is no unique answer to any of the three subparts. In particular, the list of requested quantities does not specify whether the problem uses a nearly-free-electron, tight-binding, or other model.

Conditional example: weak one-dimensional periodic potential. Suppose, in addition to the surviving text, that one is given

$$V(x) = \sum_{n\in\mathbb{Z}} V_{nG} \mathrm{e}^{\mathrm{i}nGx}, \qquad G = \frac{2\pi}{a}, \qquad V_{-nG} = V_{nG}^{*}.$$

At the first free-electron crossing, the waves $k = G/2$ and $k - G = -G/2$ are coupled by V_G. Their 2×2 Hamiltonian has equal diagonal terms at the crossing and off-diagonal terms V_G, V_G^{*}, so the leading first gap is

$$\Delta_1 \simeq 2|V_G|.$$

The crossing of $+G$ and $-G$ involves V_{2G}, hence the leading direct second gap is $\Delta_2 \simeq 2|V_{2G}|$. If a Fourier coefficient vanishes, a higher-order effective coupling can still open a gap. A zero first-order expression is not proof of an exact zero gap.

For the first crossing let $k = G/2 + q$, $B = \hbar^2/(2m)$, $K = G/2$, and $v = |V_G| > 0$. Diagonalizing the two-wave matrix gives

$$E_{\pm}(q) = V_0 + B(K^2 + q^2) \pm \sqrt{v^2 + 4B^2K^2q^2}.$$

Expanding the lower branch at $q = 0$,

$$E_{-}(q) = V_0 + BK^2 - v + \left(B - \frac{2B^2K^2}{v}\right) q^2 + O(q^4),$$

so the electron curvature mass obeys

$$\boxed{\frac{1}{m^*} = \frac{1}{\hbar^2} E_{-}''(0) = \frac{1}{m} - \frac{\hbar^2 G^2}{4m^2|V_G|}.}$$

For a weak potential the curvature is negative, as expected at the top of the lower band; the corresponding hole mass is positive. These formulas require an isolated weak-potential crossing and are not numerical answers to the cropped question.

3. One-orbital nearest-neighbor chain

Take orthonormal localized orbitals $|n\rangle$ at $x = na$, on-site matrix element ϵ_0, and real nearest-neighbor hopping $\gamma < 0$:

$$H = \epsilon_0 \sum_n |n\rangle\langle n| + \gamma \sum_n \big(|n\rangle\langle n+1| + |n+1\rangle\langle n|\big).$$

The Bloch state is $|k\rangle = N^{-1/2} \sum_n \mathrm{e}^{\mathrm{i}kna}|n\rangle$. Acting with H and relabelling the two hopping sums yields

$$H|k\rangle = [\epsilon_0 + \gamma \mathrm{e}^{\mathrm{i}ka} + \gamma \mathrm{e}^{-\mathrm{i}ka}]|k\rangle, \qquad E(k) = \epsilon_0 + 2\gamma \cos ka.$$

Since $E(0) = \epsilon_0 + 2\gamma$,

$$\boxed{E(k) = E(0) + 2\gamma(\cos ka - 1).}$$

For $\gamma < 0$, $k = 0$ is a minimum, $k = \pi/a$ is a maximum, and the bandwidth is $4|\gamma|$. Near the minimum, $E(k) = E(0) - \gamma a^2 k^2 + O(k^4)$, giving $m^* = -\hbar^2/(2\gamma a^2) > 0$.

The printed phrase "inter-orbital overlap" must mean the hopping *energy* here. A dimensionless nonorthogonal overlap $s = \langle n \mid n+1\rangle$ would instead lead to a generalized eigenproblem and, in the analogous nearest-neighbor model, $E(k) = (\epsilon_0 + 2\gamma \cos ka)/(1 + 2s \cos ka)$.

4. True or false

(a) False. A crystal can have multiple identical atoms in its basis. Diamond has an FCC Bravais lattice with two identical atoms at 0 and $a(1,1,1)/4$. Their difference is not an FCC lattice vector. Translating by that displacement does not map the whole diamond lattice onto itself. Identical chemical species therefore need not be translation-equivalent basis sites.

(b) True for a point strictly outside the first zone. By definition the first Brillouin zone is the reciprocal-space Wigner–Seitz cell

$$\mathcal{B}_1 = \{\boldsymbol{k} : |\boldsymbol{k}| \le |\boldsymbol{k} - \boldsymbol{G}| \text{ for every reciprocal vector } \boldsymbol{G}\}.$$

Failure to belong to this closed cell means that some nonzero $\boldsymbol{G}$ satisfies $|\boldsymbol{k} - \boldsymbol{G}| < |\boldsymbol{k}|$. On a zone boundary equality can occur; the statement's strict comparison concerns points outside, not boundary points.

5. FCC planes and planar density

The primitive-cell volume and reciprocal vectors are

$$\Omega = \boldsymbol{t}_1 \cdot (\boldsymbol{t}_2 \times \boldsymbol{t}_3) = a^3/4,$$
$$\boldsymbol{g}_1 = \frac{2\pi}{a}(-1,1,1), \quad \boldsymbol{g}_2 = \frac{2\pi}{a}(1,-1,1), \quad \boldsymbol{g}_3 = \frac{2\pi}{a}(1,1,-1).$$

They satisfy $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$.

(a) First direction. The stated normal is only a direction; its half-integer coefficients do not make it a reciprocal lattice vector. The shortest nonzero reciprocal vector in this direction is

$$\boldsymbol{G}_0 = \boldsymbol{g}_1 + 2\boldsymbol{g}_2 - 2\boldsymbol{g}_3 = \frac{2\pi}{a}(-1,-3,5).$$

The integer triple $(1, 2, -2)$ is primitive. For $\boldsymbol{R} = n_1\boldsymbol{t}_1 + n_2\boldsymbol{t}_2 + n_3\boldsymbol{t}_3$, the plane index is $n_1 + 2n_2 - 2n_3$, which runs through every integer. Hence successive occupied planes obey $\boldsymbol{G}_0 \cdot \boldsymbol{R} = 2\pi\ell$, and

$$\boxed{d = \frac{2\pi}{|\boldsymbol{G}_0|} = \frac{a}{\sqrt{35}}.}$$

Using 2π divided by the length of the original half-vector would incorrectly skip every other occupied plane.

(b) Density. There is one point per primitive volume, so the volume density is $1/\Omega = 4/a^3$. Dividing space into layers separated by d gives

$$\boxed{\rho_{2D} = \frac{d}{\Omega} = \frac{4}{a^2\sqrt{35}}.}$$

An independent check is the in-plane primitive basis

$$\boldsymbol{u} = -2\boldsymbol{t}_1 + \boldsymbol{t}_2, \qquad \boldsymbol{v} = 2\boldsymbol{t}_1 + \boldsymbol{t}_3.$$

Every integer solution of $n_1 + 2n_2 - 2n_3 = 0$ is an integer combination of these vectors. Their cell area is $|\boldsymbol{u} \times \boldsymbol{v}| = a^2\sqrt{35}/4$, confirming the reciprocal density.

(c) Direct-space normal. Here

$$\boldsymbol{t}_1 + 2\boldsymbol{t}_2 - 2\boldsymbol{t}_3 = \frac{a}{2}(0, -1, 3).$$

The shortest reciprocal vector along this direction is

$$\boldsymbol{G}_1 = 2\boldsymbol{g}_1 + 3\boldsymbol{g}_2 - \boldsymbol{g}_3 = \frac{2\pi}{a}(0, -2, 6).$$

Its primitive integer coefficients $(2, 3, -1)$ show that no shorter collinear reciprocal vector is allowed. Thus

$$\boxed{d_1 = \frac{a}{\sqrt{40}} = \frac{a}{2\sqrt{10}}.}$$

The vector $(2\pi/a)(0, -1, 3)$ has mixed odd/even Cartesian indices and is not a reciprocal vector of FCC.

6. Honeycomb critical points and density of states

Set $S(\boldsymbol{k}) = |F_{\boldsymbol{k}}|^2$ and $E_\pm = \varepsilon_p \pm \gamma\sqrt{S}$. Away from $S = 0$,

$$\nabla E_\pm = \pm\frac{\gamma}{2\sqrt{S}}\nabla S.$$

One must not apply this quotient at a zero of S without a local expansion.

(a) Classification. At Γ, expanding both cosines gives

$$S(\boldsymbol{q}) = 9 - \frac{3a^2}{2}(q_x^2 + q_y^2) + O(q^4), \qquad |F| = 3 - \frac{a^2}{4}(q_x^2 + q_y^2) + O(q^4).$$

Therefore

$$E_\pm(\Gamma + \boldsymbol{q}) = \varepsilon_p \pm 3\gamma \mp \frac{\gamma a^2}{4}(q_x^2 + q_y^2) + O(q^4).$$

Both bands have zero gradient. The lower band has a minimum (M_0); the upper band has a maximum (M_2), where the Morse index counts negative Hessian eigenvalues.

At $Q = (\pi/a, \pi/(\sqrt{3}a))$, write $\boldsymbol{k} = Q + \boldsymbol{q}$. The cosine factors vanish at Q, and direct expansion yields

$$S = 1 + a^2q_x^2 + \sqrt{3}a^2q_xq_y + O(q^4), \qquad |F| = 1 + \frac{a^2}{2}q_x^2 + \frac{\sqrt{3}a^2}{2}q_xq_y + O(q^4).$$

Consequently the Hessian of $|F|$ is

$$H_{|F|} = a^2 \begin{pmatrix} 1 & \sqrt{3}/2 \\ \sqrt{3}/2 & 0 \end{pmatrix}, \qquad \lambda = \frac{3a^2}{2}, -\frac{a^2}{2}.$$

In orthonormal principal coordinates (u, v) this becomes

$$E_\pm(Q + \boldsymbol{q}) = \varepsilon_p \pm \gamma \pm \frac{\gamma a^2}{4}(3u^2 - v^2) + O(q^4).$$

Thus both bands have an M_1 saddle. The mixed derivative is essential; the trace of the Hessian alone cannot distinguish a saddle from an extremum.

At $P = (4\pi/(3a), 0)$, $S(P) = 0$. Expanding S gives

$$S(P + \boldsymbol{q}) = \frac{3a^2}{4}|\boldsymbol{q}|^2 + O(|q|^3), \qquad E_\pm = \varepsilon_p \pm c|\boldsymbol{q}| + O(q^2), \quad c = \frac{\sqrt{3}\gamma a}{2}.$$

This is a Dirac contact. Each energy-ordered branch has a conical cusp, so its gradient is undefined exactly at the contact. It is not an ordinary differentiable stationary point of Morse type. The directional slopes are finite and direction-dependent; an infinite gradient inferred by manipulating $0/0$ is incorrect. The direct band gap there is zero.

Point	$\lvert F\rvert$	Lower band	Upper band
Γ	3	Minimum, M_0	Maximum, M_2
Q	1	Saddle, M_1	Saddle, M_1
P	0	Dirac contact	Dirac contact

(b) Choose the lower-band minimum at Γ. Let $E_0 = \varepsilon_p - 3\gamma$ and $\alpha = \gamma a^2/4$. Define the DOS per physical area, including g_s spin states, by

$$\frac{D(E)}{A} = \frac{g_s}{(2\pi)^2} \int \mathrm{d}^2 q \, \delta(E - E_0 - \alpha q^2).$$

For an energy close enough to the edge that the contour stays in the chosen neighborhood,

$$\begin{aligned} \frac{D_\Gamma(E)}{A} &= \frac{g_s}{(2\pi)^2} 2\pi \int_0^\infty q \, \mathrm{d}q \, \delta(E - E_0 - \alpha q^2) \\ &= \frac{g_s}{4\pi\alpha}\Theta(E - E_0) = \boxed{\frac{g_s}{\pi\gamma a^2}\Theta(E - E_0)}. \end{aligned}$$

For spin degeneracy two, this is $2/(\pi\gamma a^2)$ above the edge and zero below. The total DOS is this result multiplied by area A. Near the upper-band maximum $E_2 = \varepsilon_p + 3\gamma$, the same coefficient multiplies $\Theta(E_2 - E)$. These are local edge formulas, not constants throughout the full band.

For comparison, one Dirac valley has

$$\frac{D_P(E)}{A} = \frac{g_s|E - \varepsilon_p|}{2\pi c^2} = \frac{2g_s|E - \varepsilon_p|}{3\pi\gamma^2 a^2}.$$

Counting both inequivalent valleys adds a factor of two; one must not count six full independent valleys merely because the hexagonal zone has six corners. A saddle at Q gives a logarithmic local singularity. With $A_q = 3\gamma a^2/4$ and $B_q = \gamma a^2/4$, its leading form per saddle and per band is

$$\frac{D_Q(E)}{A} \sim \frac{g_s}{4\pi^2\sqrt{A_q B_q}} \log \frac{E_c}{|E - E_Q|},$$

where the cutoff E_c depends on the chosen neighborhood. A two-dimensional saddle is not described by the step function of a quadratic minimum.

7. Bloch theorem and its meaning

For a one-electron Hamiltonian with a Bravais-periodic potential,

$$V(\boldsymbol{r}+\boldsymbol{R}) = V(\boldsymbol{r}),$$

the lattice-translation operators commute with the Hamiltonian and with one another. The eigenstates can be chosen to be simultaneous translation eigenstates:

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}\psi_{n\boldsymbol{k}}(\boldsymbol{r}).$$

Writing $u_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\psi_{n\boldsymbol{k}}(\boldsymbol{r})$ then gives

$$\boxed{\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r}+\boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}).}$$

The label $\boldsymbol{k}$ is defined modulo a reciprocal lattice vector and may be restricted to the first Brillouin zone. Bloch's theorem does not assert that the full wavefunction is lattice-periodic, nor that the electron is a free particle; the periodic factor encodes the crystal potential.